

Methods for Platform Accountability: TikTok’s Ownership Change and Moderation of Political Content

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Abstract

Social media platforms shape political information consumption, yet platform ownership’s influence remains opaque and difficult to audit. We present a data pipeline and suite of methods for investigating changes in content moderation and algorithmic down-ranking on social media platforms. We apply these methods to the January 22, 2026 transfer of U.S. TikTok from Chinese-owned ByteDance to a U.S.-controlled joint venture. Leveraging a large-scale dataset of over 1,000,000 tracked videos and 17,924 matched TikTok-YouTube cross posts, we test two prominent hypothesis: 1) that the shift in ownership should increase the availability of anti-China content, and 2) that the transfer to Trump-allied investors would increase the demotion of liberal U.S. political content. We find no substantial evidence of changes in moderation following that ownership change at any horizon. These methods offer a replicable framework for ongoing platform accountability research.

Introduction

Social media increasingly shapes the consumption of news and information around the globe [Pew Research Center, 2025, Egan et al., 2026]. Yet, despite regulatory and civil society efforts to encourage visibility, there is very little transparency on how platform ownership influences information consumption. This creates an accountability problem where platforms can be co-opted by commercial and political interests, with potentially important implications for political outcomes such as elections and political violence [Mitts, 2025, Enikolopov et al., 2020, Gauthier et al., 2026].

One challenge is that the most consequential forms of platform intervention are the most difficult to see. Outright removal can be detected by researchers who consistently track content over time and detect disappearance [Casas, 2024]. But platforms increasingly moderate content through recommendation systems rather than deletion [Jaidka et al., 2023]. This “soft” moderation is invisible by design. Prior attempts to detect platform bias have relied on indirect proxies such as hashtag prevalence across platforms [Network Contagion Research Institute and Miller Center, Rutgers University, 2023] or likes-to-views ratios [Finkelstein et al., 2024], but these measures cannot separate platform intervention from differences in audience composition or content quality. Producing credible evidence of moderation bias requires new methods.

We develop four complementary methods for auditing platform behavior, each targeting a different layer of the moderation stack. First, we test for **selective keyword blocking** by tracking whether specific keywords disappear from posted content during alleged suppression windows. Second, we measure **content removal** by tracking which videos already posted subsequently disappear. Third, we estimate **algorithmic demotion** by using two complementary methods that identify abnormal stalls in a video’s view trajectory in the hours after posting. Finally, we measure **differential reach** by comparing views-per-followers by content group, capturing promotion and suppression effects that removal and demotion indicators may miss. We supplement each of these methods using a **matched TikTok-YouTube design** where we control for content by following the same videos posted by the same content creators before and after TikTok’s ownership change on different platforms.

Substantively, we shed light on how the TikTok ownership change affected the moderation and recommendation of political videos related to politics in China and the U.S. We test two commonly held hypotheses: 1) that the shift away from Chinese ownership of the platform should increase the availability of anti-China content and 2) that the transfer of ownership to Trump-friendly U.S. companies should increase the availability of content promoting conservative stances on politically salient issues in the U.S.

We test these hypotheses with a dataset of nearly one million tracked TikTok videos and 17,924 matched TikTok-YouTube pairs spanning the January 2026 ownership change. We do not find substantial evidence that the transfer altered moderation of either China-related or U.S. political content. If anything, contrary to the dominant narrative, pro-China content was removed and demoted by TikTok at somewhat higher rates than anti-China content both before and after the ownership change, and YouTube demoted China-sensitive content relatively more than TikTok. For U.S. political content, removal and demotion rates show no systematic ideological tilt in either direction.

Background: The TikTok Controversy

TikTok’s explosive popularity in the U.S. [Bestvater, 2024] initiated substantial concern among U.S. policymakers about foreign influence in U.S. social media [Palmer, 2022]. Whereas previously, most popular social media companies in the U.S. were U.S.-based and owned, TikTok was created and owned by ByteDance, a company headquartered in mainland China. As the PRC is a major strategic competitor to the U.S. with an extensive system of domestic online censorship and government control over digital data, policymakers’ concerns were two-fold. First, the *data* concern was that China’s National Intelligence Law requires that companies in China comply with government demands for data. Potentially, sensitive data on U.S. users could then easily be accessed by the Chinese Communist Party (CCP), and used to undermine U.S. interests. Second, the *content* concern was that China’s control over the company could allow it to censor or push content to users at the behest of China’s government, potentially changing the minds of U.S. users of the platforms. Indeed, several interviews of employees and TikTok users in 2019 and 2020 suggested that TikTok may have been moderating anti-China content [Lee, 2019, Harwell and Romm, 2019, Radio Free Asia, 2020, Hern, 2019].

These concerns have led to multiple actions by U.S. policymakers and TikTok. Following allegations in 2019 that there was pro-China censorship on TikTok’s platform, TikTok moved its content moderators outside of Beijing [Fox Business, 2020]. In November 2019, U.S. policymakers began a national security investigation into TikTok, and in November and December, the military ordered that TikTok be removed from all military personnel’s phones [Associated Press, 2024]. In 2020, President Trump issued an executive order requiring that TikTok be bought from ByteDance in 90 days. However, this executive order was blocked by the courts, and subsequently dropped at the beginning of the Biden administration [Allyn, 2020].

In 2021, the Biden administration began negotiations with ByteDance called “Project Texas” through

the Committee on Foreign Investment in the United States (CFIUS) to resolve the issue by routing all U.S. TikTok data through Oracle servers in the United States [Binkley, 2024]. These talks failed, however, in part because of reports that there was still China-based access to U.S. user data despite TikTok’s assurances to the contrary [Baker-White, 2022]. Thus, bipartisan political will began building in Congress to ban TikTok. In 2024, Congress passed the Protecting Americans from Foreign Adversary Controlled Applications Act (PAFACA) requiring TikTok be sold, or otherwise be banned in the U.S. In 2025, the Supreme Court upheld the ban, though the Trump administration paused enforcement repeatedly in order to negotiate an ownership transfer.

The Trump administration negotiated the separation of the U.S. version of the TikTok app from the ByteDance version of the TikTok app, effective January 22, 2026. The U.S. version of the app is now 50% controlled by a combination of Oracle, Silver Lake, and MGX with ByteDance holding less than a 20% stake. Importantly, any U.S. data under the deal is held on U.S. servers and the U.S. algorithm is retrained by the new entity on U.S. data [Duffy, 2025]. Even though the U.S. users use TikTok in the same app as global users, their content moderation and algorithm as of January 2026 are both controlled by the U.S. entity, and not ByteDance.¹ The January 22, 2026 date therefore creates a natural event to study whether ownership influences platform behavior.

The composition of TikTok’s new ownership group raised a second concern. The primary U.S. investors including Oracle, Silver Lake, and MGX are run by some executives with close ties to the Trump administration, and the deal itself was negotiated under Trump’s direct involvement [Zahn, 2026]. This has led people to worry that content moderation on TikTok could serve the interests of Trump and the Republican party, similar to what scholars and pundits believed to have happened after Elon Musk’s takeover of X [Gauthier et al., 2026]. Whether a comparable shift occurred on TikTok after the January 2026 transfer is the second question this paper addresses.

Despite this national-level controversy and legislation, there is very little quantitative empirical research on whether or to what extent the TikTok’s platform put its thumb on the scale to support pro-PRC content or in how it has changed its algorithm since the sale of U.S. TikTok. While before the takeover there were important anecdotal instances of unjustified removal of anti-PRC content, it is unclear whether these instances were reflective of platform-wide policies, or are merely isolated errors. Existing platform-wide empirical work examining whether TikTok had a pro-China bias before the takeover focused on the prevalence of anti-PRC hashtags on TikTok, comparing them to other platforms such as Instagram [Harwell, 2019,

¹<https://www.forrester.com/blogs/tiktok-seals-the-deal-with-new-us-joint-venture/>

[Network Contagion Research Institute and Miller Center, Rutgers University, 2023](#)]. These studies find that anti-China hashtags were more prevalent on non-TikTok platforms like Instagram than they were on TikTok. While this is suggestive of manipulation, many have pointed out that this could be driven by other factors such as audience engagement, and not platform moderation policies themselves.

There are two notable exception to this. First, [Lin \[2021\]](#) conduct a security and privacy analysis of TikTok in comparison to Douyin. Despite the fact that TikTok has a “sensitive” category in its source code that could be used to flag sensitive political content, they find inconclusive evidence that TikTok is engaged in political censorship and show that unlike Douyin, TikTok does not constrain politically sensitive terms in search. Second, [Finkelstein et al. \[2024\]](#), examine engagement with pro- and anti-China content on TikTok in comparison to YouTube before the takeover. They find that anti-China TikTok videos on topics such as Tibet, Tiananmen, and Xinjiang have significantly higher likes to views ratios than pro-China videos on the same topic. This could indicate that despite more engagement (higher likes) with that content, it is less promoted by the platform (leading to fewer views). They also find that this ratio is higher on TikTok than on YouTube, suggesting that TikTok may have been downgrading anti-China content despite its popularity. While this evidence is more suggestive than previous work that the platform could be intervening, it is still possible that it reflects different audience engagement patterns and different content posted across platforms.

More generally, the paucity of large-scale empirical studies of TikTok’s moderation of content reflects the larger problem of auditing social media companies’ treatment of user-generated content. Despite significant public controversy about the platform’s policies, it is difficult for third parties to audit the moderation, demotion and promotion of content on social media platforms because of lack of data access to researchers [\[Persily and Tucker, 2020\]](#). While policymakers and civil society have made increasing calls for more platform transparency in content moderation [\[Suzor et al., 2019\]](#), general reticence on part of platforms has made work in this area – even when it is central to national security – close to impossible.

Data Sampling Strategy

Our findings rely on four main samples of interest:

1. U.S. TikTok accounts
2. A snowball sample of TikToks posted from accounts that focus on China
3. A sample of U.S. accounts focused on U.S. politics

4. A joint TikTok-YouTube sample of identical videos from (2) and (3) posted on both platforms

Dataset 1: Sample of all U.S.-based TikTok accounts

Our first sample consists of a large random sample of TikTok videos posted by users located in the United States. Following the approach described by [Steel et al. \[2025\]](#), we continuously identified newly uploaded posts in near real time and collected their metadata, including the account’s country and a content category field, which together allowed us to isolate political TikTok videos produced in the US. The resulting collection spans US political TikTok production from July 2024 to July 2026, with an estimated overall coverage of roughly 70%: about seven in ten US TikTok videos appear in our dataset, with no detectable selection bias. For each video, we re-collected metadata 30 days after upload, allowing us to determine whether the video had been removed and to measure the engagement it generated, including the full set of comments it received. The resulting dataset contains 5.6 billion observations, 40 million of which are political.

Dataset 2: Snowball sample of international producers focusing on Chinese politics

We collect a sample of China-related TikTok accounts, which we follow closely for a 1.5 year time frame. To do this, we first use TikTok search to find videos associated with China-related keywords and identify a seed sample of accounts that post China-related content. To be sure that we identify accounts from multiple perspectives, we collected 113 primarily political accounts posting favorable content toward the CCP, 125 primarily entertainment accounts posting favorable content toward the CCP, 63 primarily political accounts posting critical content toward the CCP, and 9 primarily entertainment accounts posting critical content toward the CCP. The remaining seed accounts either lacked a clear stance label (58, mostly political accounts), were state-media outlets (3), or lacked a clear content-type label (12). In total, we collected 383 accounts as our seed sample.

We then expand our collection of accounts as follows. We use Dataset 1 to identify 701,845 videos authored by our seed accounts to identify all accounts ($N = 6,664,197$) that commented at least once on videos authored by our seed accounts – we call these the seed sample’s followers. Then, we identified all additional accounts that produced at least one piece of content commented on by five or more of the seed sample’s followers. These are likely to be popular accounts in the same topic area as the original seed accounts. We then tasked two AI classifiers to identify, based on the description of each account’s 10 most-commented videos, which of the 215,076 accounts were actually primarily posting “about China.” We

hand-verified those accounts that had inconsistent labels between the classifiers. We ended with a list of 4,700 popular, China-related accounts.

Because of data collection constraints, we identify 2,000 of these 4,700 accounts our classifiers deemed most likely to be about China. For these accounts, in addition to the initial seed accounts, we return to the account every 30 minutes to check for new videos, which we download as soon as we see them. For each new video, we check the status of and view counts of the video for each 30 minutes for three days after it was initially posted online. From November 2024 through June 2026, we collected 970,156 videos in the China sample and, from November 2025 through June 2026, 637,882 videos in the U.S. sample. For the China sample, continuous full-panel collection began on July 2025; we therefore restrict the China analysis 763,362 videos posted between July 2025 and June 2026, the period over which each video is observed for at least its first 72 hours. Brief collection interruptions in mid-March and on May 5, 2026 leave a handful of post-dates in the sample without full forward coverage; we drop these days from removal- and demotion-sensitive analyses.

Dataset 3: A sample of U.S. accounts focused on U.S. politics

Our third sample follows prominent U.S. accounts that focus on U.S. politics. Starting from the most prominent U.S. political TikTok accounts identified in Dataset 1, we rank accounts by follower count and retain the top 2,500 English-language accounts, supplementing them with a hand-curated list of media outlets and politicians. For these accounts we mirror the collection design of Dataset 2: we revisit each account every 30 minutes to detect newly posted videos, and we then track each new video’s availability and view counts every 30 minutes over its first three days, with an additional revisit at seven days to measure longer-horizon removal. This collection yields 637,882 videos posted between November 2025 and June 2026, of which 629,749 fall on days with a complete 72-hour observation window.

Dataset 4: Joint TikTok-YouTube Sample

We identify authors in Dataset 2 and 3 who also have an account on YouTube, and track engagement with their YouTube content in a very similar fashion: we check newly posted videos every 30 minutes, and measure engagement metrics for each new video every 30 minutes for the first three days, with an additional seven-day check. This matched sample allows us to assess the extent to which content curation and moderation on TikTok differs from other social media platforms. To identify those who are also on YouTube, we look for whether a YouTube channel is mentioned in the profile description of the TikTok accounts: 631 of the

4,700 China-related accounts in Dataset 2, and 626 of the 2,500 US-related accounts in Dataset 3. We began tracking these China-related channels on February 2025, and the US-related ones on November 2025, for a total of 330,508 and 339,884 YouTube videos in total, respectively.

Presence on two platforms does not necessarily imply the same activity on both of them. For a more stringent identification, we identify instances where content creators posted the same video on both platforms. For each potential pair of YouTube-TikTok videos that we have collected from each author, we calculated the cosine similarity between the ngram vectorization of the text we had available for each video in the pair: title and description for the YouTube videos, and description, hashtags, and voice-to-text for the TikTok videos. We consider a pair of TikTok-YouTube videos to be a match if their cosine similarity score is above 0.7. We validated the performance of this classification approach on a set of 122 video pairs that we manually annotated: 84% accuracy, 92% precision, and 67% recall. If a YouTube video is matched to more than one TikTok from the same author, we only take the most similar match into account. In total, we are able to match 17,924 TikTok-YouTube videos: 10,696 for the China-related collection (Dataset 2) and 7,228 for the US-focused accounts (Dataset 3).

Methods for Detecting Moderation

Our audit strategy works inward from the most visible forms of intervention to the least. **Keyword blocking** and **content removal** are observable in principle. If a video disappears, or all posts of a particular keyword vanish, this should be observable to both creators and consumers. **Algorithmic demotion** by contrast is more invisible. If a video remains online but stops being recommended, the consumer and creator receive very little signal. **Differential reach** captures the net effect of promotion and suppression for content that continues to circulate.

All four methods share a common identification strategy: we compare content across content groups (by stance toward China or by political ideology) before and after the January 22, 2026 ownership change. In addition to looking at the full sample, for content removal, algorithmic demotion, and differential reach, we use TikTok-YouTube pairs of identical videos to separate out platform-specific behavior from content characteristics and creator behavior.

Selective Keyword Blocking

The most basic allegation of content moderation is selective keyword blocking. In selective keyword blocking, companies block social media posts that contain a particular keyword. If such blocking is occurring at scale, we should see no posts containing a particular keywords during the time period of keyword blocking.

TikTok has previously been accused of selective keyword blocking. First, following the TikTok ownership change in late January, 2026, there were allegations that TikTok was selectively preventing the upload of content related to ICE following the shooting of Alex Pretti.² TikTok’s response was that its data centers had been affected by an outage on those days due to a large storm that affected much of the U.S.³

Second, TikTok has been accused of selectively blocking particular keywords at the direction of the CCP. For example in 2023, *Forbes* reported that TikTok’s parent company ByteDance had a sensitive word list that included options to “kill” posts containing those keywords. While TikTok denied allegations that the lists were relevant to them, some of the keyword lists contained TikTok’s name in them explicitly [Levine, 2023].

To study keyword blocking we look at the numbers of posts with sensitive political keywords in our largest dataset (Dataset 1) over time. For the China sample, we investigate keyword occurrences of Xi Jinping and June 4. For the U.S. sample, we look at “ice,” “alex pretti,” “renee good,” “trump,” and “epstein.” For comparison purposes, we look at placebo words in the same time period. For China, we examine “Dragon Boat Festival” (which also occurs in late May/early June) and basketball player “Yao Ming.” For the U.S., we compare the prevalence of sensitive keywords to placebos “recipe” and “oscar.”

Content Removal

Next, we investigate claims of changes in **content removal**. Revisiting videos that we have previously scraped to see if they are still posted, we look at post deletion – initiated either by the user or the platform. We do this for Datasets 2-4 to understand whether or not patterns of post deletion change systematically across relevant content categories before and after the ownership change. Because neither TikTok nor YouTube lets us distinguish creator-initiated deletion from platform takedowns, we interpret *differences* in removal across content groups, platforms, and time — not removal levels — throughout.

We measure removal at three horizons. Our primary measure is a *3-day removal* indicator: a video is

²<https://www.nytimes.com/2026/01/27/us/tiktok-ice-minneapolis-technical-problems.html>

³<https://usdsjv.tiktok.com/Technical-issues-impacting-TikTok-US-service>

considered removed if its last successful observation falls short of 71.5 hours after posting.⁴ We complement it with a *7-day removal* indicator from the seven-day revisit. Finally, for the part of the sample in the U.S. and China datasets that overlaps with Dataset 1, we add a longer-horizon *30-day removal* indicator which is a revisit of these videos thirty days after they were posted. This thirty-day measure is available only for the videos in Dataset 1 (a random sample of the labeled videos), so all 30-day rates are computed over that sample and videos without a 30-day measure are removed from 30-day analyses.

Video Demotion

We develop two new methods for estimating **demotion** of social media posts – where a post initially retains a lot of views, and then is algorithmically downranked to receive very few further or zero additional recommendations in user feeds, and thus few additional views. In contrast to removal, demotion (aka downranking or shadowbanning) cannot be directly observed. Our measurement strategy defines a downranking indicator by observing an abnormal temporal pattern in which a post initially accumulates views rapidly but then abruptly stalls. We call this “flat-lining” and focus on instances most suggestive of a platform intervention. Past political communication work has modeled decay curves in sentiment dials while participants watch political ads [Iyengar et al., 2016], though we are not aware of other efforts to model longitudinal social media engagement curves to study algorithmic speech moderation.

Gompertz Model

Our first measure of video demotion comprises two key steps:

- Step 1: Using a modified Gompertz model trained on video- and account-level covariates, we estimate the fraction of cumulative engagement over time that we would expect given the author’s prominence and other video meta-data.
- Step 2: We compare the actual observed temporal pattern to that which we would expect under the model, allowing us to assess how much faster engagement plateaus than expected.

The model used in the first step estimates the expected cumulative fraction of views over time using a modified Gompertz function (this model has a rich history modeling growth curves in biology, see for example Tjørve and Tjørve [2017]). This model flexibly captures the extent to which views accumulate early

⁴This is defined only for cohorts old enough to be fully observed and set to missing on the small number of posting days affected by collection outages (the 72-hour follow-up coverage on those days is too low to read removal reliably).

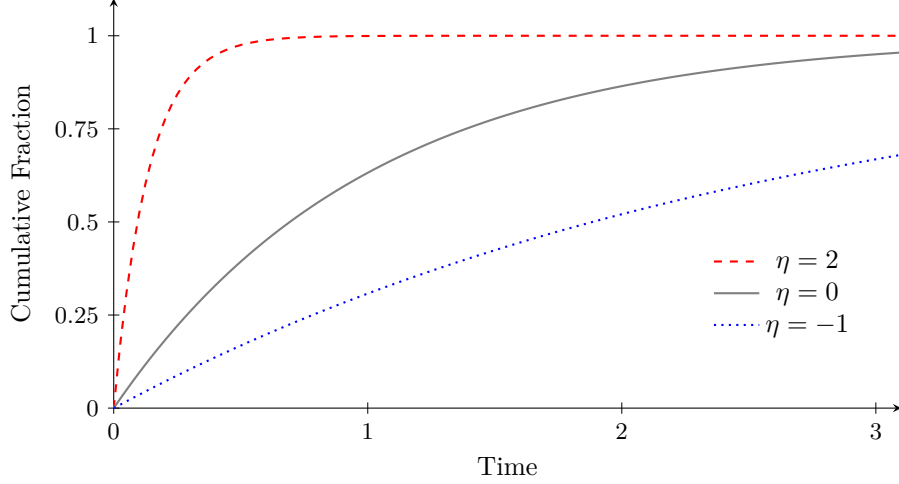


Figure 1: Modeling the cumulative fraction of views using the Gompertz function, using a set of linear predictors represented by η_i , over time. Higher values of η_i correspond to a relatively early rise and sudden pause in view accumulation.

and attenuate over time, unlike symmetric growth functions (e.g., logistic). This allows us to capture the early-peaking view patterns of interest.

Formally, let y_{it} denote the cumulative fraction of views for video i in time bin t . We model y_{it} as:

$$\hat{y}_{it} = 1 - \exp(-\exp(\eta_i) \cdot t) \quad (1)$$

$$\eta_i = \mathbf{X}\beta \quad (2)$$

Where t is the index of the time bin (30-minute intervals since posting) and η_i is a linear combination of video-level covariates that affect the rate and shape of the growth. Each coefficient in β vector controls the influence of a particular video attribute on the shape and timing of the view trajectory. Higher values of η_i correspond to a relatively early rise followed by a sudden pause – “early flat-lining” – in view accumulation.

The model is fit using nonlinear least squares (NLS) via the `nlsLM` function from the `minpack.lm` R package. The time series is binned into 30-minute intervals up to a maximum duration, and the model is fit to predict the fraction of total 72-hour views observed up to each bin.

We model the expected cumulative fraction of views conditional on the account’s follower count (strongly associated with total reach) and the video’s accumulated likes and comments as of the previous time bin, which capture the video’s own early engagement trajectory.⁵

⁵All three covariates enter as lagged running counts, i.e., their values as of the previous 30-minute observation bin.

Then, we take the mean over residuals, r_i (the average actual - expected cumulative views) and form an indicator variable (EF_i) to form our “early flat-lining” variable to study the phenomenon of interest. We use the percentile of EF_i to determine whether early flat-lining is unusual. For example, an above 80th percentile early-flatlining video would be defined as:

$$\bar{r}_i = \frac{1}{T} \sum_{t=1}^T (y_{it} - \hat{y}_{it}) \quad (3)$$

$$EF_i = \mathbb{1}\{\bar{r}_i \geq \hat{q}_{0.8}\} \quad (4)$$

The 80th-percentile indicator above is the original exploratory cutoff. In the analyses in this paper we work with the percentile *rank* of $\bar{r}_i$ directly (the “early-peaking score”) and form demotion indicators at fixed selectivity cutoffs — the top 1%, 5%, 10%, and 25% of videos by score — matching the cutoffs used for the change point model below, so that the two demotion models always flag the same share of videos and can be compared directly.

Figure 2 presents examples of individual temporal trajectories and the associated early flat-lining percentile. Observed cumulative views (black) are shown against the model-predicted baseline (red dashed) over the first 72 hours. High-percentile cases (top row) display a rapid initial rise followed by an abrupt, sustained plateau that remains well below the baseline. This is our “early flat-lining” pattern, that we deem consistent with a platform intervention. Mid-percentile cases (middle row) track the baseline closely, showing no suspicious flattening. Low-percentile cases (bottom row) instead start slowly and accelerate later, ultimately matching or exceeding the baseline.

Change Point Model

Our second measure takes a complementary, model-free approach. Rather than comparing a video’s trajectory to a fitted growth curve, it asks whether the trajectory contains a single abrupt break in view *velocity* — the signature of a video that circulated normally and then suddenly stopped being recommended.

For each video we construct an hourly panel of cumulative views over its first 72 hours: within each hour-since-posting bin $t \in \{0, 1, \dots, 71\}$ we take the last observed cumulative view count and carry the last observation forward across unobserved hours. Videos whose first scrape *observation* arrives after their first hour — 5.5% of U.S. and 4.6% of China videos, a property of scraper timing rather than of the video’s views — are treated as *unassessable*: with the start of the trajectory unobserved, an early break cannot be

Observed vs. Predicted Views for Sample TikToks

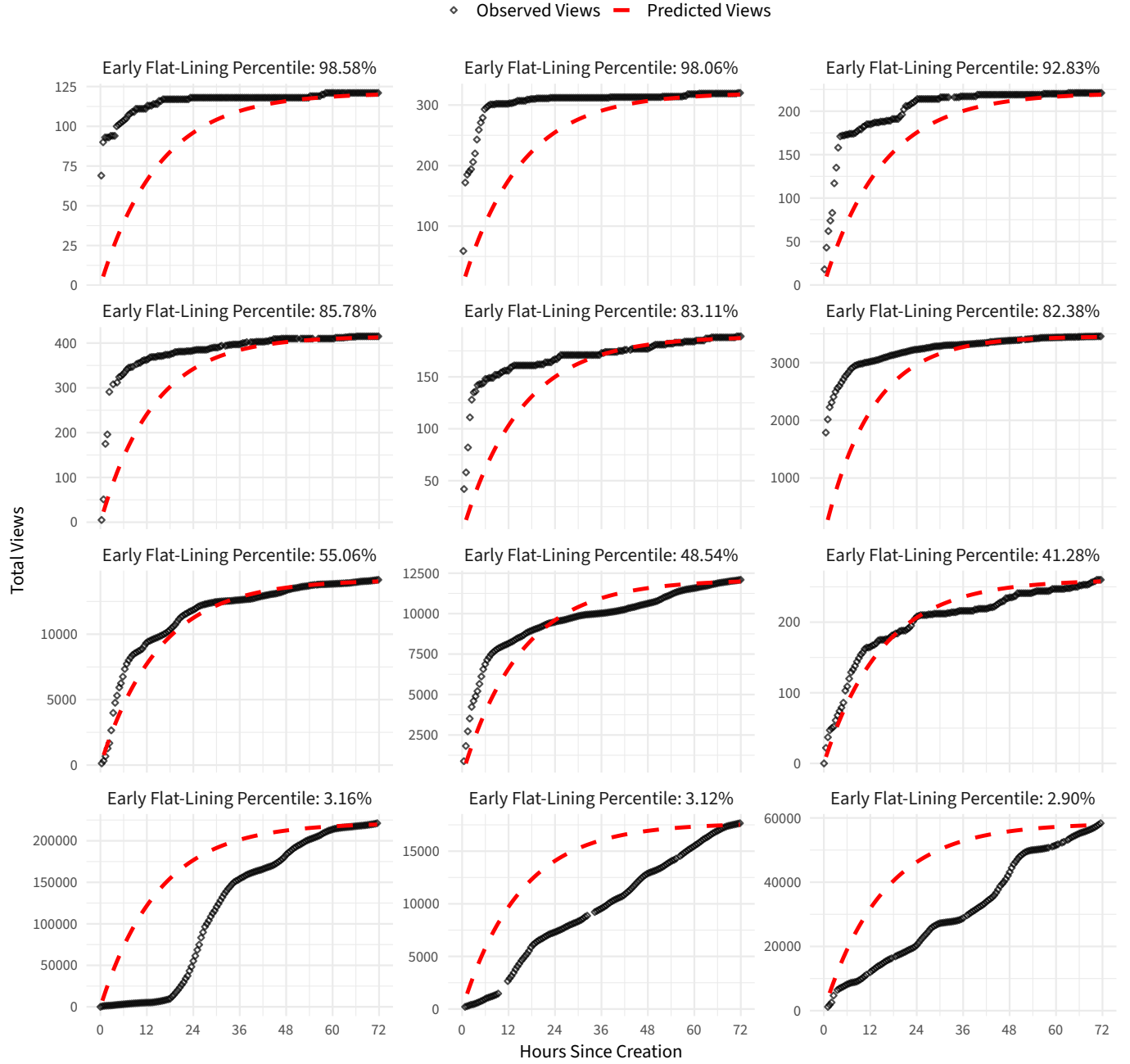


Figure 2: Predicted vs. observed views for a sample of TikTok videos, along with our early flat-lining percentile measure.

distinguished from late observation.⁶ Let $C(t)$ denote the views accumulated by the end of hour t , so that

⁶Unassessable videos receive no demotion score rather than being classified as not demoted; they still enter the removal measures, and in the union suppression measure introduced below a removed-but-unassessable video counts as suppressed. This anchoring requirement is not consequential: anchoring instead at the first observation within three hours and carrying that

$C(72)$ is the video’s total 72-hour views. For every candidate change point $\tau \in \{6, 7, \dots, 66\}$ — requiring at least six hours of data on each side — we compare the mean hourly view velocity before and after τ :

$$\bar{v}_{\text{pre}}(\tau) = \frac{C(\tau)}{\tau}, \quad \bar{v}_{\text{post}}(\tau) = \frac{C(72) - C(\tau)}{72 - \tau}, \quad \rho(\tau) = \frac{\bar{v}_{\text{pre}}(\tau)}{\max\{\bar{v}_{\text{post}}(\tau), 0.5\}} \quad (5)$$

where the denominator is floored at 0.5 views per hour so that videos whose circulation stops entirely do not produce divide-by-zero artifacts. The estimated change point is $\hat{\tau} = \arg \max_{\tau} \rho(\tau)$. This is equivalent to fitting a two-level step function to the hourly view velocity and locating the break that maximizes the pre/post drop; the video’s demotion score is the drop ratio at that break, $\rho(\hat{\tau})$. Figure 3 illustrates the detector on two stylized trajectories: a throttled video (a steep climb followed by near-zero velocity) produces a tall, sharp peak in $\rho(\tau)$ at the break, while a video with sustained reach produces a low, flat profile.

A natural flagging rule is an absolute threshold on the drop ratio (e.g., flag when $\rho(\hat{\tau}) > 10$ with a minimum pre-break velocity). In our data such absolute rules are far too permissive: the threshold rule above flags roughly 39% of all U.S. videos — implausibly many to constitute deliberate platform intervention — because the attention paid to most short-video content naturally decays steeply within hours of posting. We instead convert the continuous drop ratio into demotion indicators at fixed *selectivity*: we flag the top 1%, 5%, 10%, and 25% of assessable videos by $\rho(\hat{\tau})$, the same percentile cutoffs we use for the Gompertz residual measure, so that the two models flag identical shares of videos and can be compared directly.

Comparing Gompertz and Change Point Models

We do not have a ground truth measure of demotion, so we cannot directly evaluate which method performs better. What we want to measure is an unusual view curve in which many views accumulate in the first few hours after a video is posted and then the video suddenly receives very few subsequent views. We therefore characterize what each measure captures by validating both against observable proxies in the U.S. sample.

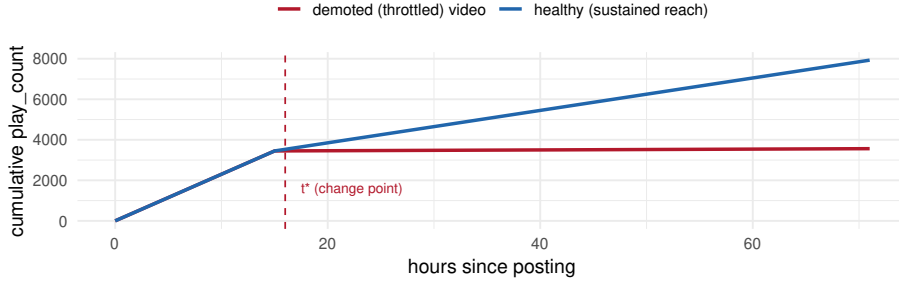
The two measures turn out to be complementary rather than redundant. First, at matched selectivity they flag largely different videos: at the top-10% cutoff, only about 9,700 of the roughly 55,000 Gompertz-flagged and 43,000 change-point-flagged videos overlap (Jaccard index ≈ 0.11 ; $\phi \approx 0.12$). Second, the Gompertz measure aligns far better with a literal “flat-lining” trajectory proxy — a sustained run of near-zero view growth between hours 4 and 72 with a saturation guard: the area under the ROC curve (AUC) against this

value backward (the rule we use for YouTube, whose first snapshot typically arrives after an hour) recovers about a fifth of the excluded videos, leaves the drop ratio of jointly assessable videos essentially unchanged (rank correlation 1.00; top-10% flag agreement above 99.9%), and flags the recovered videos at slightly *below* the nominal rate (7.8% and 9.0% at the top-10% cutoff), indicating that the exclusion does not censor a demotion-prone subpopulation. The remaining excluded videos have their first observation later than three hours after posting; no anchoring rule can assess an early break for them.

Change point demotion detector: a single break in view velocity

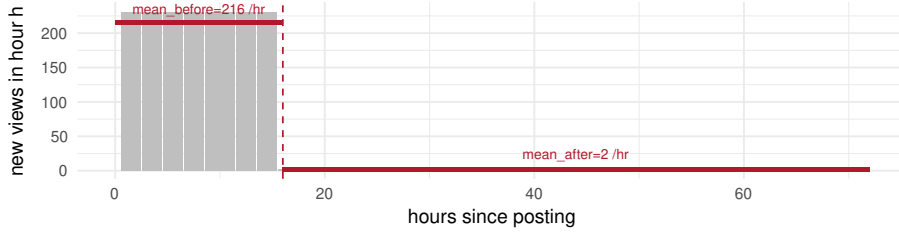
A. Cumulative views $V(h)$ over first 72h

The throttle = a sharp KINK: steep climb then near-flat



B. Hourly velocity $\Delta V(h)$ (throttled video)

The detector fits a two-LEVEL step function to the velocity



C. Drop-ratio profile $\rho(t)$ over candidate change points

$t^* = \text{argmax } \rho(t)$. Throttle $\rightarrow$ tall sharp peak; saturation $\rightarrow$ low/broad

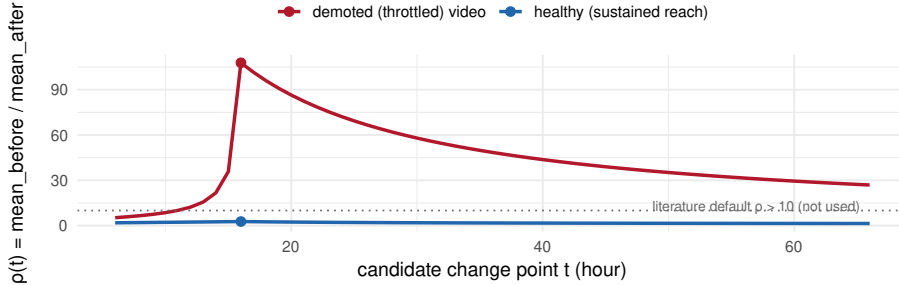


Figure 3: The change point demotion detector, illustrated on two stylized videos. Panel A: cumulative views over the first 72 hours. The throttled trajectory (red) shows the sharp kink the detector targets. Panel B: the detector fits a two-level step function to the hourly view velocity of the throttled video, with segment means before and after the estimated change point $\hat{t}$. Panel C: the drop-ratio profile $\rho(\tau)$ over candidate change points. The throttled video produces a tall, sharp peak at the break, while the sustained-reach video stays low. The dotted line marks the fixed threshold $\rho > 10$ used in absolute flagging rules, which we reject in favor of selectivity-matched percentile cutoffs.

proxy is 0.81 for the continuous early-peaking score and 0.70 for the top-25% flag, versus essentially chance (0.50–0.54) for the change point flags at any cutoff. Third, the change point measure is instead *removal-adjacent*: videos in its top 1% by drop ratio are roughly 13 times more likely to be subsequently removed within 7 days (48% vs. 3.7% removal among unflagged videos), whereas Gompertz flags carry almost no information about subsequent removal (risk ratios at or below 1).

Substantively, the Gompertz model detects *throttling* — trajectories that stall early relative to what the video’s own starting conditions predict — while the change point model detects *abrupt cutoffs* that often precede outright removal. In the analyses that follow we therefore report demotion using both models at matched selectivity, treat their *agreement* (a video flagged by both models) as our most conservative demotion indicator, and define overall suppression as the union of 3-day removal and both-model agreement, computed on the common set of videos for which both models are assessable.

Differential Reach

To evaluate the overall reach of videos and accounts by category before and after the ownership change, we use two methods. First, we evaluate capture the total views for a video after three days. To take into account number of followers, we also measure total views for a video after three days divided by the total number of followers of the account. While differential reach can reflect a lot of different components of the algorithm, it is a useful metric is estimating whether different communities and topics’ audiences changed following the transfer of ownership.

Labeling

In order to assess whether removal, demotion, and reach, are impacted by type of content, the last component we need before analysis is video and account-level labeling. For the China and U.S. datasets, we take two slightly different approaches. For the China dataset, we label accounts at the video level. In contrast, because U.S. social media accounts generally have consistent ideological lean, we label the U.S. dataset at the account level.

Video-level Labeling

For the China-related analysis, we label at the video level based on two sets of categories: (1) Community-Guidelines-based categories and (2) China-related categories. To assess whether TikTok’s content removal primarily affects certain types of content, we label each video using both human coders and large language models (LLMs).

First, following TikTok’s Community Guidelines, we label whether a video is likely to violate TikTok’s content rules. Under TikTok’s own policies, such content should be removed or demoted. We use these labels to validate our measures of removal and demotion. We distinguish between four different potential

guideline violations.⁷

- **Misinformation:** Content that contains misinformation or conspiracy theories
- **Hate speech:** Attacks, discrimination, dehumanization based on group attributes (race, ethnicity, gender, sexual orientation, religion, disability, etc.), promotion of hateful groups, bullying, degrading statements, or coordinated harassment
- **Harmful content:** This include nudity, violent content, harmful or dangerous activities, youth exploitation, or regulated substances.
- **News Clips:** This is where > 25% of video comes from news program(s), which are potentially affected by algorithmic demotion.

Second, we label a set of categories, that capture how the content relates to China.

- **China-related** Whether or not the video is related to China or Taiwan
- **Stance toward the PRC Government (`stance_prc_gov`):** Evaluates whether the content supports, praises, criticizes, or mocks the CCP, PRC government, leadership, or political system, including stances on Taiwan unification and minority regions.
- **Stance toward Chinese Culture (`stance_culture`):** Evaluates whether the content portrays Chinese citizens, traditions, cuisine, festivals, and ethnic minorities in a positive, negative, or neutral light.
- **Stance toward Chinese Technology (`stance_tech`):** Evaluates whether the content praises or criticizes mainland Chinese technology, innovation, companies, or infrastructure engineering.
- **China-sensitive issues:** This includes the treatment of politically sensitive topics in China in a way that deviates from official line of the CCP, including Taiwan independence, pro-DPP party content, issues related to political repression in Hong Kong, the independence of Tibet, the repression of Uyghurs, the June 4 movement, criticisms of Xi Jinping or the CCP.
- **Collective action** Whether or not the video is about a protest, or otherwise encourages or portrays collective action.

⁷An earlier labeling pass additionally distinguished *inauthentic* and *derivative* content; the revised prompt (July 2, 2026) consolidates the misinformation dimension into its own category and drops those two. All label-based China-sample results in this draft use the revised labels.

Constructing a Labeled Dataset

Because removal and demotion are rare events and the sample exhibits severe class imbalance, we pursued a stratified sampling approach for human labeling. We partitioned our video population into three equally sized strata: early flat-lining (top percentile of from our Gompertz model), mid stratum (2nd to 98th percentile) and late blooming (bottom percentile). We then oversampled for removed videos within each stratum, aiming for an even split between removed and non-removed videos where possible. In total, we selected 10,000 videos for screening.

We then labeled this sample of 10,000 with Gemini 2.5, a popular vision-language model (VLM): We submitted each video together with our code book and constrained predicted tokens to follow our coding scheme. We then selected 900 total videos to attempt to obtain class balance: 20% of the sample were random, 40% were rated by Gemini to have a non-neutral value for China-related stance detection, and 40% were rated by Gemini to have a non-neutral value for violating community guideline variables. We asked two research assistants to each code 500 videos, 111 of which overlapped. See table 1 for the inter-coder reliability results.

Table 1: Inter-Rater Reliability (IRR) Results across Classified Variables

Variable	Krippendorff’s α	N pairs
collective_action	0.938	111
harmful	0.826	111
hate_speech	0.826	111
stance_culture	0.617	61
stance_gov	0.578	61
news	0.569	111
china_related	0.545	111
sensitive	0.340	61
stance_tech	0.332	61

Gemini Model

We again labeled the sample of 900 human coded videos with Gemini, asking it to provide a short free-form text to justify its labeling choices. An initial investigation of performance metrics comparing human labeling with Gemini labeling showed that model performance was seemingly poor (see table 2).

Three members of the research team and a highly trained human coder then systematically inspected cases of disagreement for each variable and found that our reviewing project members agreed with the VLM’s coding choices in the majority of cases. Based on the reviewed videos, the VLM outperforms human coders

Table 2: Macro F_1 Performance Across Classified Variables

Variable	Macro F_1 (Coder 1)	Macro F_1 (Coder 2)
china_related	0.493	0.379
collective_action	0.367	0.371
harmful	0.301	0.321
hate_speech	0.321	0.372
news	0.469	0.508
sensitive	0.389	0.494
stance_culture	0.363	0.428
stance_gov	0.327	0.377
stance_tech	0.468	0.379

because it exceeds them in terms of attention span, attention to detail and knowledge of political actors, events, and locations that are shown in videos. This out-performance was especially true for China-related and stance variables, where Gemini significantly outperformed the human coders. Human coders were in contrast more accurate for harmful and hate speech variables, where Gemini was likely to over-categorize videos as harmful or hateful. In light of our findings, we revised the Gemini codebook to correct the hate speech and harmful inaccuracies. For the remainder of the paper, we use this revised Gemini model to categorize the variables.

Account-level Labeling

For this U.S. sample ($N = 2,440$), in May 2026 we manually annotated accounts by their political ideology on a five-point scale: Extremely Liberal ($N = 169$), Liberal ($N = 708$), Moderate-Neutral ($N = 369$), Conservative ($N = 341$), and Extremely Conservative ($N = 101$); plus a non-political category for those accounts that in the end did not produce politically-relevant content ($N = 593$). We combine the separately coded *Moderate* ($N = 67$) and *Neutral* ($N = 302$) accounts into a single Moderate-Neutral category; for further simplification, in some of the subsequent analysis we also collapse the extremely liberal and liberal, and extremely conservative and conservative, categories. We were unable to annotate the remaining accounts in this Dataset 3 ($N = 159$) because they had no videos, or had been removed or turned to private by the time we performed this ideological annotation. Two research assistants worked on all these annotations. They initially agreed 47% of the time (65% when collapsing the extremely liberal and liberal, and extremely conservative and conservative, categories), for a two-rater weighted Cohen’s kappa of 0.44. In a second round, the two annotators discussed the instances in which they disagreed, achieving a consensus annotation for all of them.

Results

Keyword Blocking Results

Using Dataset 1, we test for allegations of **selective keyword blocking**. First, we investigate the claim that TikTok was keyword blocking at the end of January 2026 by looking at the overall amount of content on the platform with the keywords “ice” and “pretti” for the time period of the block. To test TikTok’s claim that the inability to upload posts was due to a server outage, in addition we tracked the posts that included more innocuous keywords such as “recipe” and “oscar.” As we show in Figure 4, we find that the pattern of content on the platform related to these words is more consistent with a platform outage than with selective keyword blocking. While there is a sharp decline in posts about ice and pretti, there is a similar decline for recipe and oscar, indicating that all posts – not just a particular political subset – were affected.⁸

While our data does not reach back to 2023, we investigate keyword blocking of sensitive China-related keywords on TikTok over the past two years. As reported by *Forbes*, one sensitive keyword on the 2023 list was 六四 (“6 4”), which refers to the date June 4, the anniversary of the crackdown on protesters in Tiananmen Square in 1989. We look at whether we see TikTok videos mentioning this keyword in the caption, particularly on anniversaries of the crackdown. We also examine whether we see consistent numbers of TikTok posts mentioning China’s president Xi Jinping.

As shown in Figure 5, we see that there are TikTok posts discussing both June 4 and Xi Jinping, consistently throughout the time period. Posts with June 4 peak on the date of the anniversary, consistent with expectations. Therefore we see no evidence of wholesale keyword blocking.

When comparing the volume of content of posts related to Xi Jinping to posts about the Chinese basketball player Yao Ming, we see that posts about Xi Jinping are considerably more common. However, when comparing the volume of content of June 4 compared to other similarly timed events, such as the Dragon Boat Festival, we do see relatively few posts on TikTok about June 4. The relative paucity of discussion about June 4 does not necessarily provide evidence of platform intervention, as there are many explanations for this including user interest and selection into TikTok. However, we cannot rule out other forms of moderation or blocking of particular posts related to these topics. We turn next to studying evidence of moderation.

⁸<https://goodauthority.org/news/was-there-censorship-on-tiktok-after-the-u-s-takeover/>

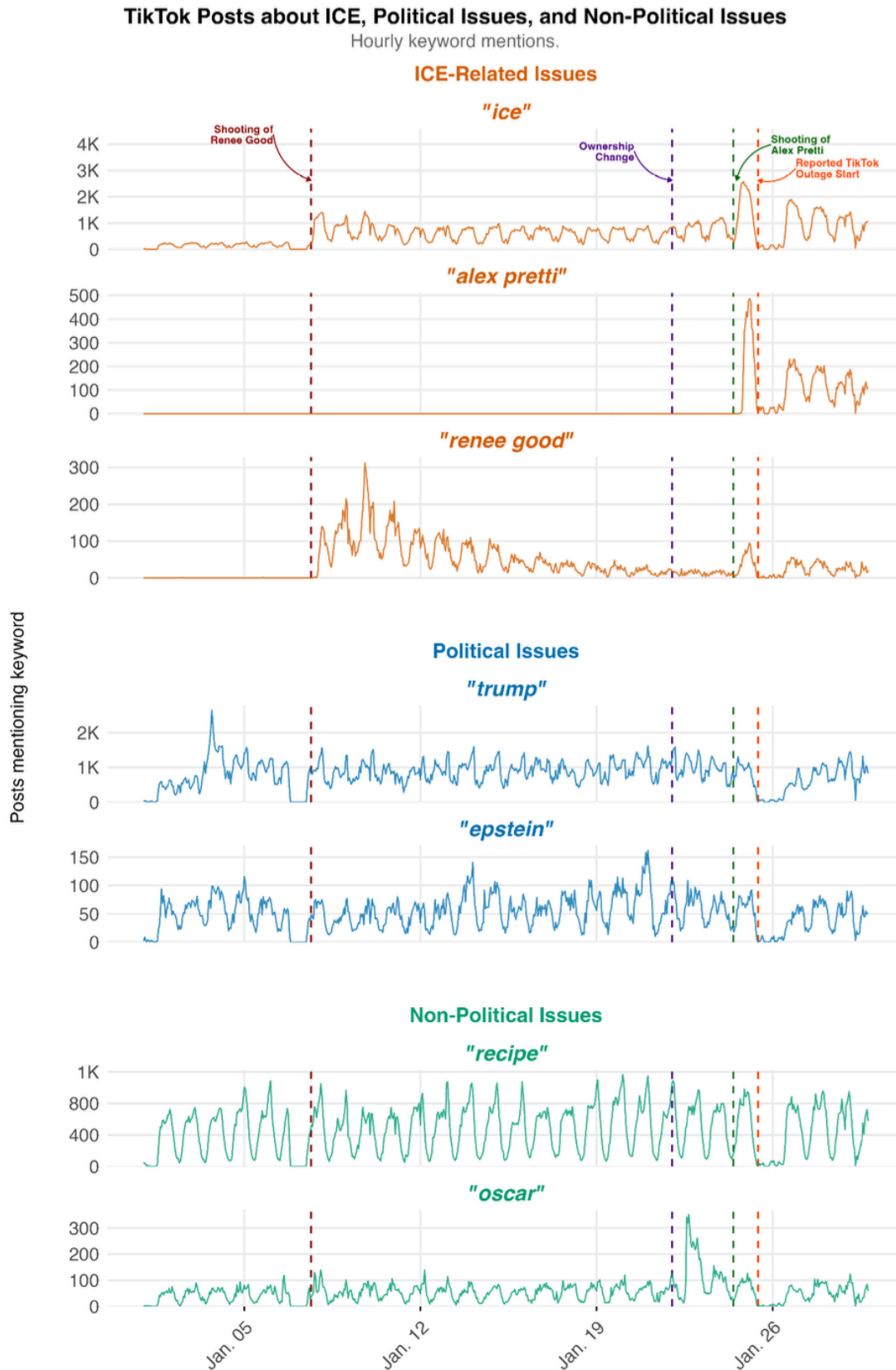


Figure 4: Hourly number of TikTok posts (Dataset 1) mentioning ICE-related keywords (“ice,” “alex pretti,” “renee good”), political keywords (“trump,” “epstein”), and non-political keywords (“recipe,” “oscar”) in January 2026. Vertical lines mark key events, including the ownership change and the start of the TikTok outage. Posting volume declines during the outage window for political and non-political keywords alike, a pattern more consistent with a platform-wide outage than with selective keyword blocking.

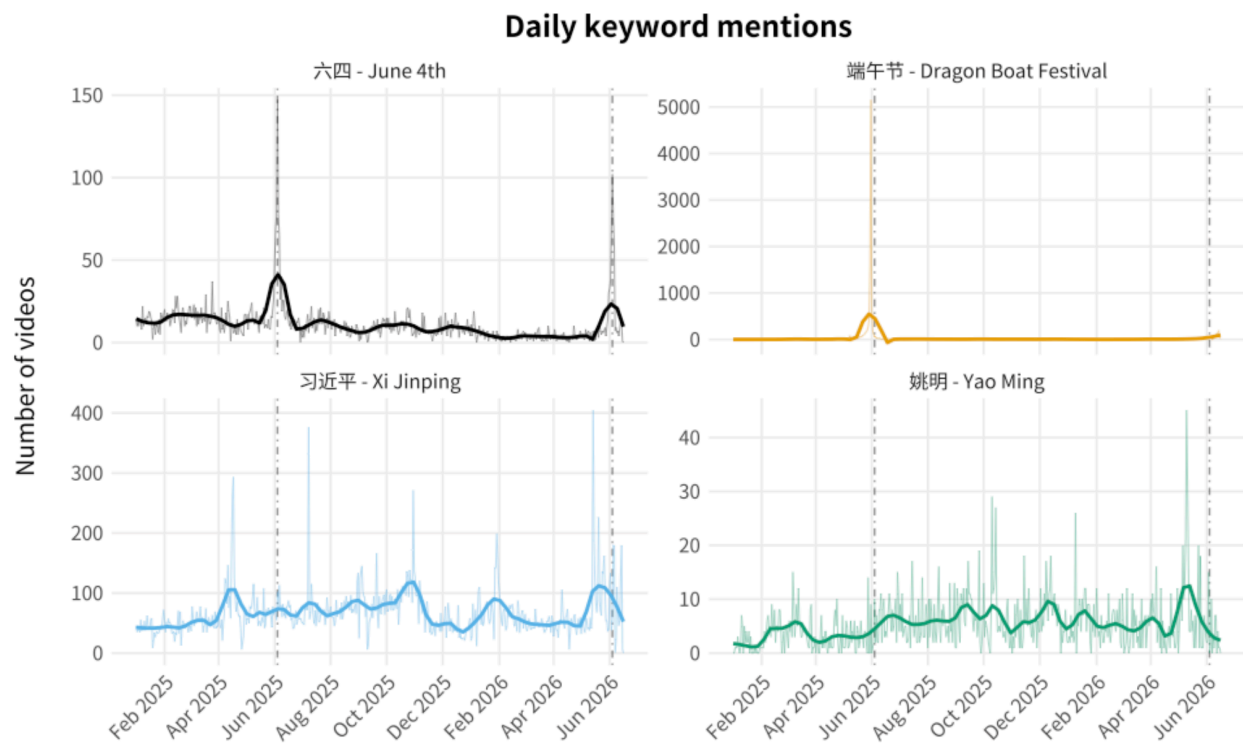


Figure 5: Daily number of TikTok videos (Dataset 1) whose captions mention 六四 (June 4), 端午节 (Dragon Boat Festival), 习近平 (Xi Jinping), and 姚明 (Yao Ming), February 2025 through June 2026. Mentions of June 4 peak on the anniversary of the Tiananmen crackdown in both years, and mentions of Xi Jinping appear consistently throughout the period, providing no evidence of wholesale keyword blocking.

Removal Results: China-Related Posts

Are anti-China posts more likely to be removed in our data? Does this change after the transfer of ownership in January 2026? We use a random sample of 12,000 posts created between 2 months before and after the ownership change from our dataset to investigate this claim.⁹ After excluding posts without videos, as described above, we submitted the remaining videos to Gemini 2.5 for labeling (10,599 labeled videos under the revised July 2 prompt).

First we look overall at the removal rate by category. Figure 6 shows results of predicting removal rate using the video-level category indicator. We see some expected patterns. First, harmful content predicts removal, particularly at seven and thirty days (+3.3 and +3.7 percentage points). Misinformation, by contrast, does not predict removal at any horizon.¹⁰ Interestingly, the news clips and collective action categories predict that the content is less likely to be removed.

For China-related categories, the results are contrary to our expectations built upon popular discussion. While China-related content is more likely to be removed in the 3 and 7-day checkpoints, pro-China content is overall more likely to be removed than anti-China content.

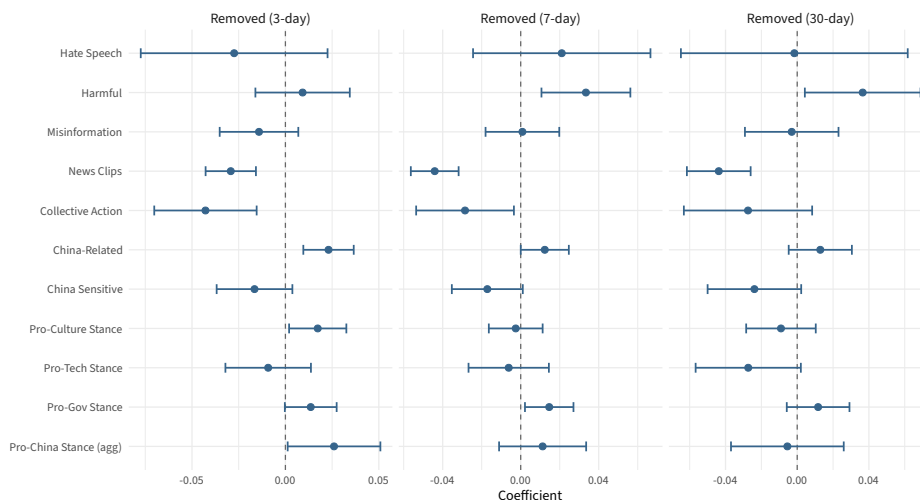


Figure 6: Bivariate associations between Gemini content labels and removal in the China sample (10,599 labeled videos posted within two months of the ownership change). Each row within a panel is a separate regression of removal (within 3, 7, or 30 days) on a single label (linear probability models); points are OLS coefficients with 95% confidence intervals.

Of course, categories can be correlated with each other. It could be that pro-China content also tends to

⁹In future iterations, we plan to classify all posts.

¹⁰Hateful content does not either; we expect that this is because Gemini has a more expansive definition of hateful than TikTok’s community guidelines – we are working at addressing this.

be more likely to violate the community guidelines. To address this, in Figure 7 we estimate the coefficient for predicting removal including all categories in the same multivariate regression (blue line). We also include author fixed effects, in case certain authors are more likely to be removed for other reasons outside of platform policies (orange line). Overall, we find similar results, with pro-China content being slightly more likely to be removed than anti-Chian content, except for the pro-tech category which is less likely to be removed at the 7 and 30 day thresholds.

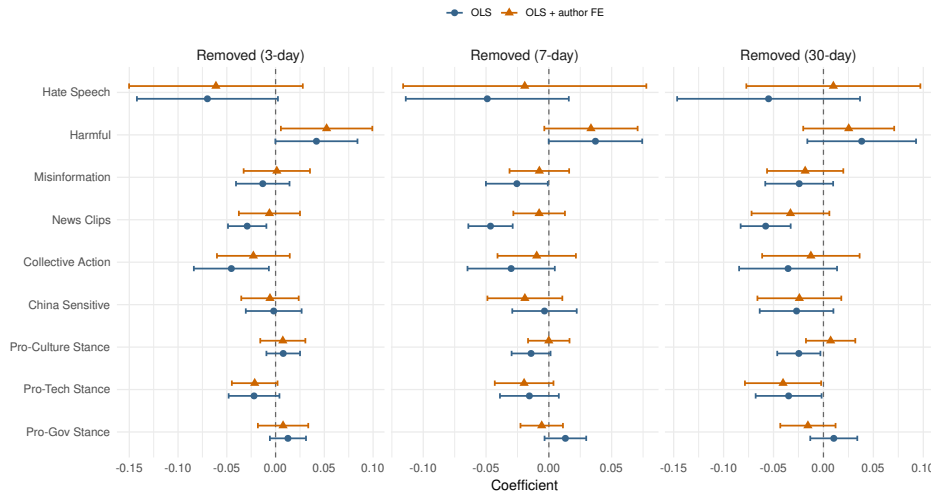


Figure 7: Multivariate associations between Gemini content labels and removal in the China sample (3-, 7-, and 30-day horizons), estimated among China-related videos only. All labels enter the model jointly, with the three community-guidelines flags (hate speech, harmful, misinformation) entered separately and stance toward China entered as its three dimensions (government, technology, culture). Blue circles: OLS; orange triangles: OLS with author fixed effects (within-creator comparisons). Bars are 95% confidence intervals.

The ownership change of U.S. TikTok was in part driven by worries about CCP influence over content moderation on TikTok. After the ownership change, do we see changes in how the stance of a video toward China is related to removal rates? We investigate this using the same Gemini-labeled sample of 12,000 posts (10,595 labeled videos) drawn from the two months before and after the ownership change, described above.

First, we look at overall videos by stance before and after the ownership change. As we can see in Figure 8, pro-China videos are the most common on TikTok in our sample, both before and after the ownership change. We do not see an increase in overall posting of anti-China videos after the ownership change, which we might expect if there was an expectation among users that content moderation rules would change.

In Figure 9, we show the removal rate by stance before and after the ownership change, at the 3-day, 7-day, and 30-day horizons. We see no difference in removal rate by category, and no difference before versus after the change, at any horizon.

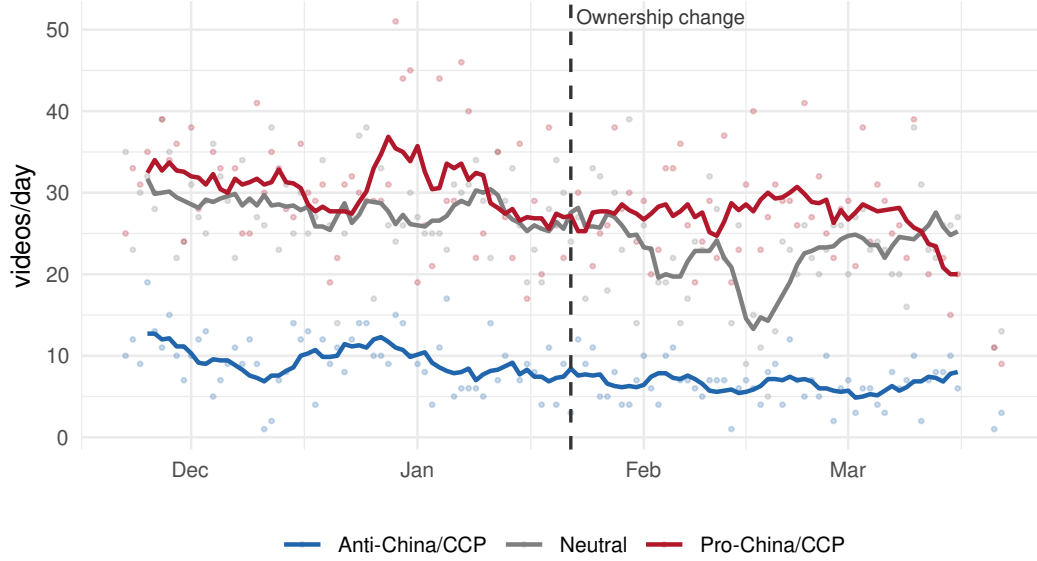


Figure 8: Daily number of videos posted in the labeled China sample by aggregate stance toward China (pro, neutral, anti), November 22, 2025 through March 22, 2026. A video’s aggregate stance averages its Gemini stance labels toward the PRC government, Chinese technology, and Chinese culture. Points are daily counts; solid lines are centered 7-day rolling means; the dashed vertical line marks the ownership change (January 22, 2026). Days with severe collection outages (under 50% 72-hour coverage) are omitted and appear as gaps.

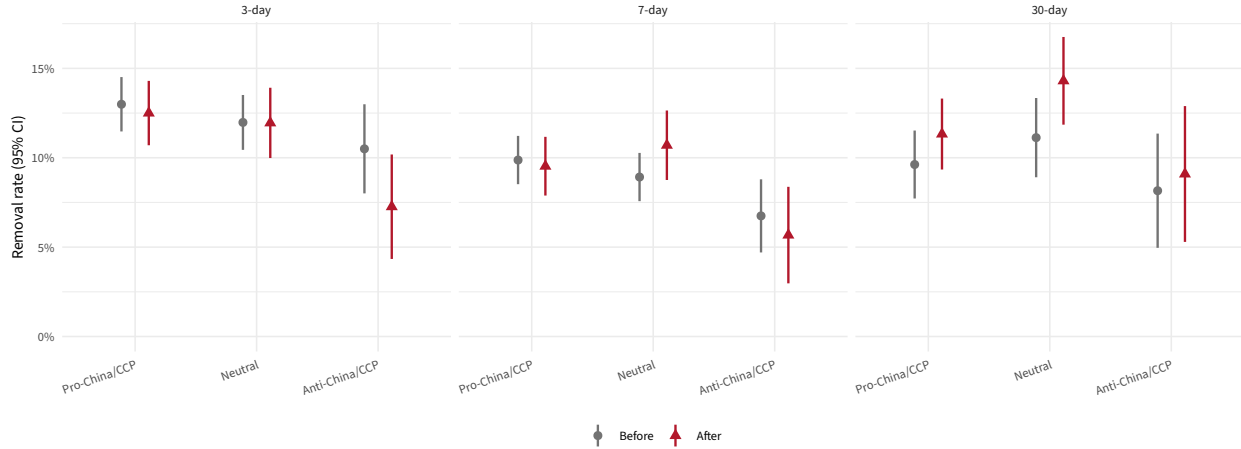


Figure 9: Removal rates in the labeled China sample by aggregate stance toward China, pooled before vs. after the ownership change, at the 3-day, 7-day, and 30-day horizons. Points are pooled removal rates with 95% confidence intervals (grey = before, red = after). Anti-China videos are not removed at systematically higher rates than pro-China videos either before or after the ownership change, at any horizon.

Last, we use the matched TikTok–YouTube sample to compare removal of identical China-related videos across platforms, using YouTube — which did not change ownership — as a baseline. To enable this comparison we extended the Gemini labeling pass to *all* matched China videos (10,299 of 10,705 pairs

labeled, under the revised July 2 prompt). Like the overall sample, pro-CCP videos outnumber anti-CCP videos (1,065 vs. 237 stance-labeled pairs, with 5,076 neutral), and 527 pairs touch China-sensitive topics — imbalances to keep in mind when reading the group contrasts below. Match propensity itself also varies with stance: Appendix Figure 27 regresses whether a TikTok video is matched to a YouTube counterpart on the posting account’s stance toward the CCP and anti-China videos are somewhat less likely to be matched.

We estimate the same pair fixed-effects specifications. Formally, for video i in matched pair $p(i)$ posted on platform $k \in \{\text{TT}, \text{YT}\}$, we estimate a pair fixed-effects specification:

$$\text{Removed}_{ik} = \alpha_{p(i)} + \beta \text{TT}_k + \gamma (\text{TT}_k \times \text{Pro-CCP}_{p(i)}) + \varepsilon_{ik} \quad (6)$$

where $\alpha_{p(i)}$ are pair fixed effects and standard errors are clustered by creator. Because a pair’s creator ideology is constant within the pair, its main effect is absorbed by $\alpha_{p(i)}$: β is the within-pair TikTok–YouTube removal gap for anti-CCP videos, and $\beta + \gamma$ is the same gap for pro-CCP videos. To test whether the pro-CCP differential changed at the ownership change, we add interactions with a post-takeover indicator (defined by the pair’s TikTok posting date, restricting to pairs cross-posted within seven days of each other), yielding a triple-difference specification:

$$\begin{aligned} \text{Removed}_{ik} = & \alpha_{p(i)} + \beta \text{TT}_k + \gamma (\text{TT}_k \times \text{pro-CCP}_{p(i)}) + \delta (\text{TT}_k \times \text{Post}_{p(i)}) \\ & + \lambda (\text{TT}_k \times \text{pro-CCP}_{p(i)} \times \text{Post}_{p(i)}) + \varepsilon_{ik} \end{aligned} \quad (7)$$

where the lower-order pro-CCP, Post, and pro-CCP $\times$ Post terms are again absorbed by the pair fixed effects. The coefficient of interest, λ , measures whether pro-CCP videos became relatively more (or less) likely to be removed on TikTok than on YouTube after the takeover, compared with anti-CCP

Figure 10 shows the proportion of matched videos removed on each platform by stance and sensitivity, before and after the ownership change. We measure removal at a horizon-matched window on both platforms: the 30-day status re-check on TikTok is paired with a YouTube removal read over the same 30-day window, defined as a video- (or whole-channel-) suspension within 30 days of the YouTube post, from a suspension log with per-video unavailability timestamps.¹¹ Because fewer than 10 pro- and anti-CCP TikTok removals fall into the stance-labeled subset, we do not have much power for this analysis; the better-powered comparison

¹¹We use the suspension log rather than a single point-in-time snapshot precisely because it lets us read each video over the same 30-day window from its own posting date, rather than over the varying window between posting and one pull date. YouTube videos posted fewer than 30 days before the July 3, 2026 pull are censored. We report the 3-day-suspension version as a robustness check (Appendix Table 5 and Figure 28).

is the China-sensitive vs. other split. Pooling periods, the pair fixed-effects stance interaction (Table 3, column 3) is +1.6 percentage points and statistically indistinguishable from zero, and the point estimate again indicates that, if anything, it is *pro*-CCP content that is removed relatively more on TikTok than on YouTube — the opposite of the hypothesis that TikTok disproportionately removes anti-CCP content.

After the ownership change, the figure shows somewhat higher YouTube suspension for anti-CCP pairs (13% vs. 9% on TikTok, among 45 pairs), but this rests on only six YouTube suspensions concentrated in a handful of creators, so the creator-clustered standard errors are wide and the corresponding triple-difference estimate (Table 3, column 4: +7.8 percentage points, $p = 0.33$) is not distinguishable from zero. We find no robust evidence that the ownership change shifted the cross-platform removal differential by stance toward China, at either the 30-day or the 3-day horizon.

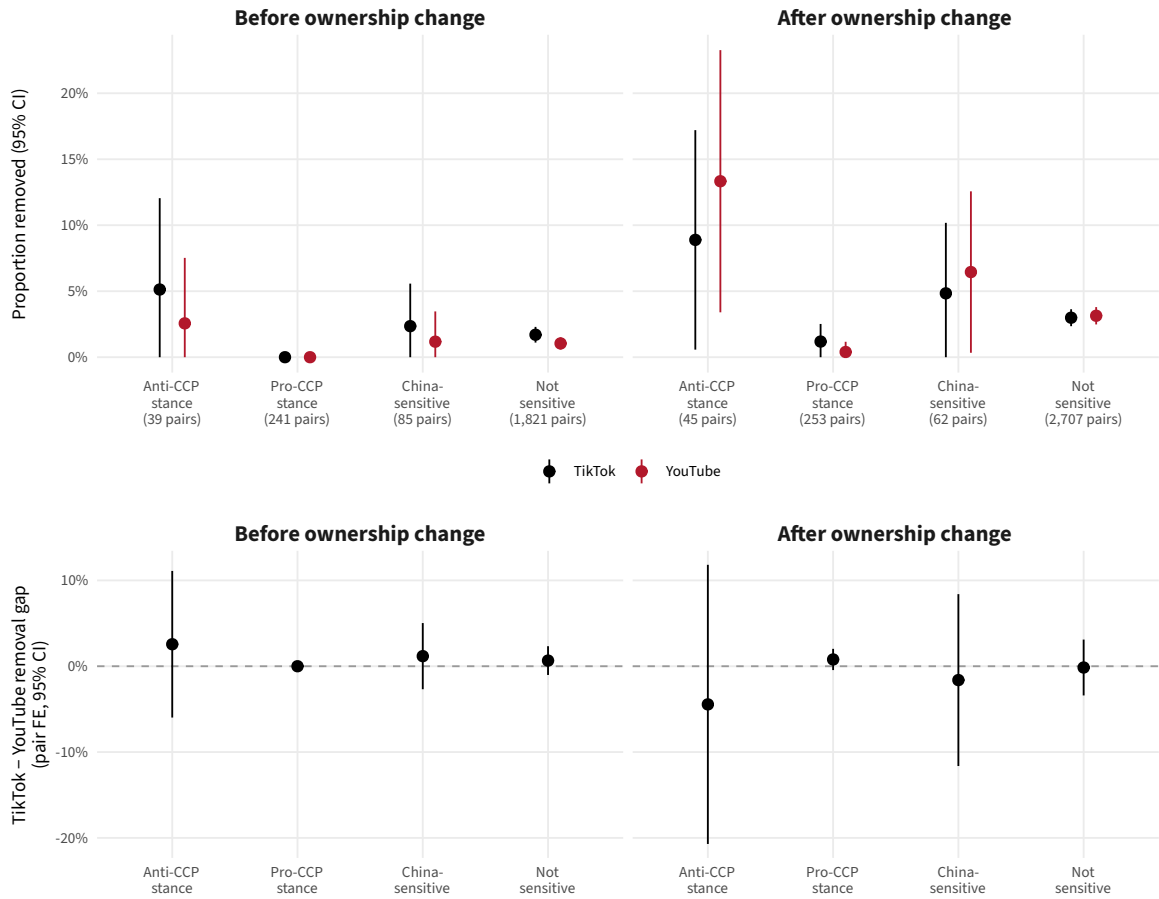


Figure 10: Removal of matched cross-posted China-sample videos on TikTok vs. YouTube by stance toward the CCP and sensitivity, for pairs posted before vs. after the ownership change (date range: February 2025–May 2026). Top: proportion removed with 95% confidence intervals. Bottom: within-pair TikTok–YouTube removal gap from a pair fixed-effects regression with creator-clustered 95% confidence intervals. YouTube removal is a video- or whole-channel suspension within 30 days of the YouTube post (matched to TikTok’s 30-day status re-check).

Removal Results: U.S.-Related Posts

The ownership change did not influence post removal by China stance, but did the change impact post removal by stance in U.S. politics? In this section, we study how the ownership change coincides with a change in removal of posts by account ideology in our Dataset 3.

First, we examine overall trends in video creation by account ideology. Figure 11 shows that liberal videos are the most common in our sample, followed by neutral and conservative videos. Very liberal and very conservative videos are the least common. Posting volume rises across all ideology groups in early January 2026—peaking around January 6–9 at roughly 20% above the December average before easing after the ownership change—so the early-January bulge in the liberal series reflects a broad, platform-wide increase (accentuated by a dip in posting over the late-December holidays) rather than a liberal-specific surge or a few high-volume accounts: the ten most active liberal accounts contribute under 15% of liberal videos even on the peak day.

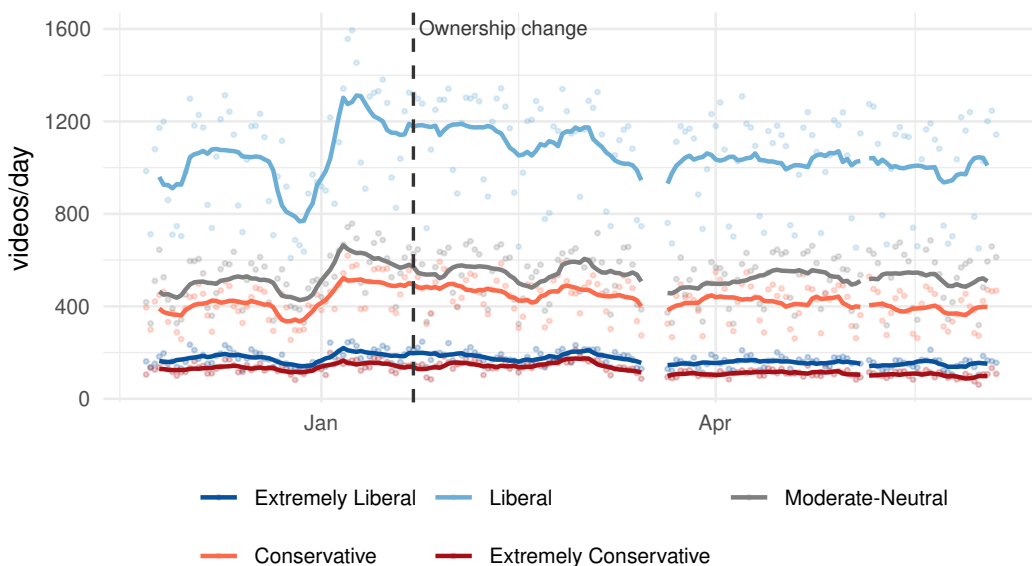


Figure 11: Daily number of videos posted in the U.S. sample (Dataset 3) by account ideology, November 22, 2025 through early June 2026. Points are daily counts; solid lines are centered 7-day rolling means; the dashed vertical line marks the ownership change; the late-March 2026 collection outage appears as a gap.

Figure 12 shows removal rates by account ideology before and after the ownership change. Compared with December, we do see higher rates of removal for extremely liberal and extremely conservative accounts relative to liberal, conservative, and neutral accounts. Removal rates are particularly high for extremely conservative accounts during the month of March. Overall, we do not see evidence that liberal accounts

have increased removal rates, as we might have expected from popular rhetoric. The longer-horizon 30-day measure (panel (b), available for the labeled status-check sample) tells the same story.

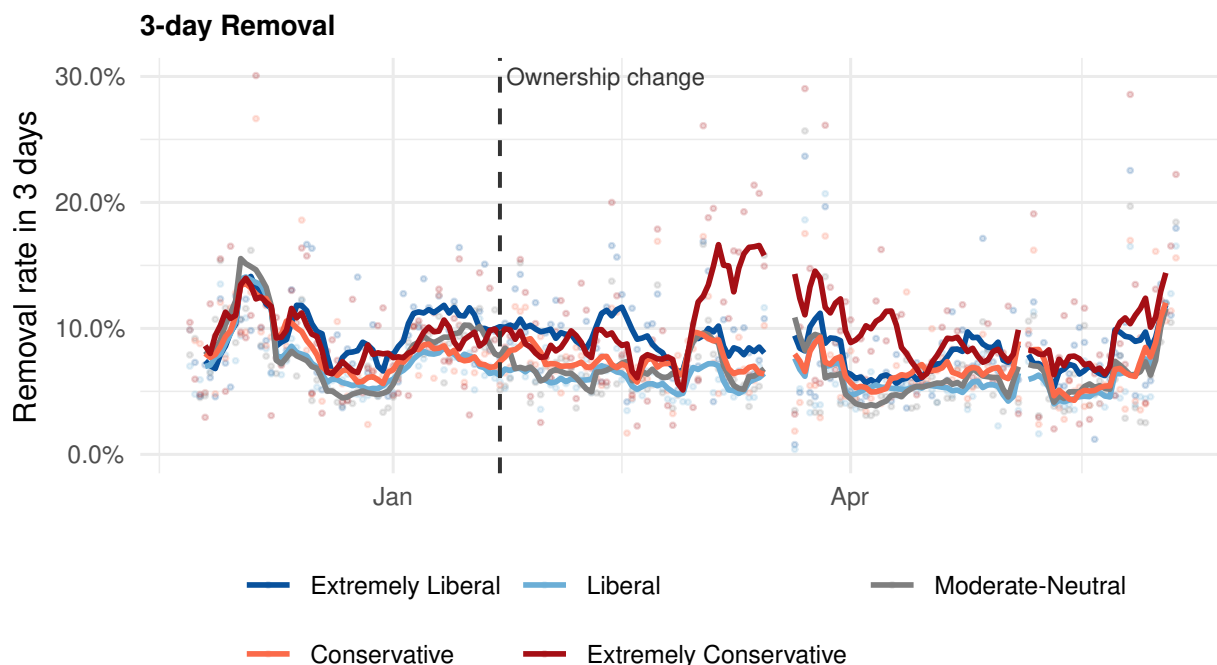


Figure 12: Removal rates in the U.S. sample by account ideology, around the ownership change (dashed vertical line). Daily 3-day removal rate (faint points are raw daily rates; solid line is the 7-day pooled rolling rate). Days within collection outages are omitted and appear as gaps. Outside of an increase in extremely conservative removal in March 2026, no ideological group shows a systematic post-takeover increase in removal.

Did the ownership change alter how TikTok moderates U.S. political content relative to other platforms? We use the matched TikTok–YouTube sample to compare removal of *identical* videos posted by the same creators on both platforms, before and after the ownership change. Because each video in a matched pair carries the same content and the same creator to both platforms’ moderation systems, the within-pair comparison nets out content characteristics, creator characteristics, and topical composition; YouTube, which did not change ownership during this period, serves as the comparison platform.¹² In the matched U.S. sample, liberal creators outnumber conservative creators roughly 3.5 to 1 (3,745 vs. 1,069 pairs, with 2,390 pairs from accounts without an ideology score), an imbalance to keep in mind when reading the ideology contrasts. Match propensity, by contrast, does not vary systematically with ideology (Appendix Figure 26).

We use the same specifications as above (Equations 6 and 7) to estimate the paired fixed-effect and triple difference specifications, swapping out pro-CCP for Conservative in each case. We use the 30-day status

¹²This assumes not large shifts in YouTube moderation policy over this time period.

check because it is most comparable to YouTube’s unavailability measure. We report the 3-day TikTok measure as a robustness check in the Appendix.

Figure 13 shows the results descriptively for the two months before and after the ownership change. At the horizon-consistent 30-day measure, the within-pair TikTok–YouTube removal gap is small and statistically indistinguishable from zero overall, and it shows no ideological tilt. Table 3 reports the corresponding estimates (columns 3 and 4 report the parallel China-sample estimates discussed in the previous subsection). In the pooled difference-in-differences (column 1), the within-pair gap is almost zero. In the triple difference (column 2), the post-takeover change in the conservative differential is $\lambda = -0.9$ percentage points ($p = 0.60$). If anything, conservatives’ content became slightly *less* likely to be removed on TikTok relative to YouTube after the takeover, but the estimate is not statistically distinguishable from zero. We include robustness checks for the removal analysis and event studies to examine parallel trends in the Appendix.

Table 3: Matched YouTube–TikTok removal (30-day status check). For the U.S. sample, “Group” is Conservative creators. For the China sample, “Group” is pro-CCP stance videos. Columns (1)/(3): pair fixed-effects difference-in-differences; columns (2)/(4): triple difference adding the post-takeover interaction. All columns restrict to pairs cross-posted within seven days. Standard errors are clustered by creator in parentheses.

	US (Group = Conservative)		China (Group = Pro-CCP)	
	(1) DiD	(2) Triple diff.	(3) DiD	(4) Triple diff.
TikTok	0.008 (0.004)	-0.004 (0.014)	-0.012 (0.055)	0.026 (0.043)
TikTok $\times$ Group	0.005 (0.007)	0.013 (0.015)	0.016 (0.056)	-0.026 (0.043)
TikTok $\times$ Post		0.015 (0.014)		-0.070 (0.079)
TikTok $\times$ Group $\times$ Post		-0.009 (0.017)		0.078 (0.079)
Observations	6,080	6,080	1,156	1,156
Matched pairs	3,040	3,040	578	578
Pair fixed effects	Yes	Yes	Yes	Yes

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

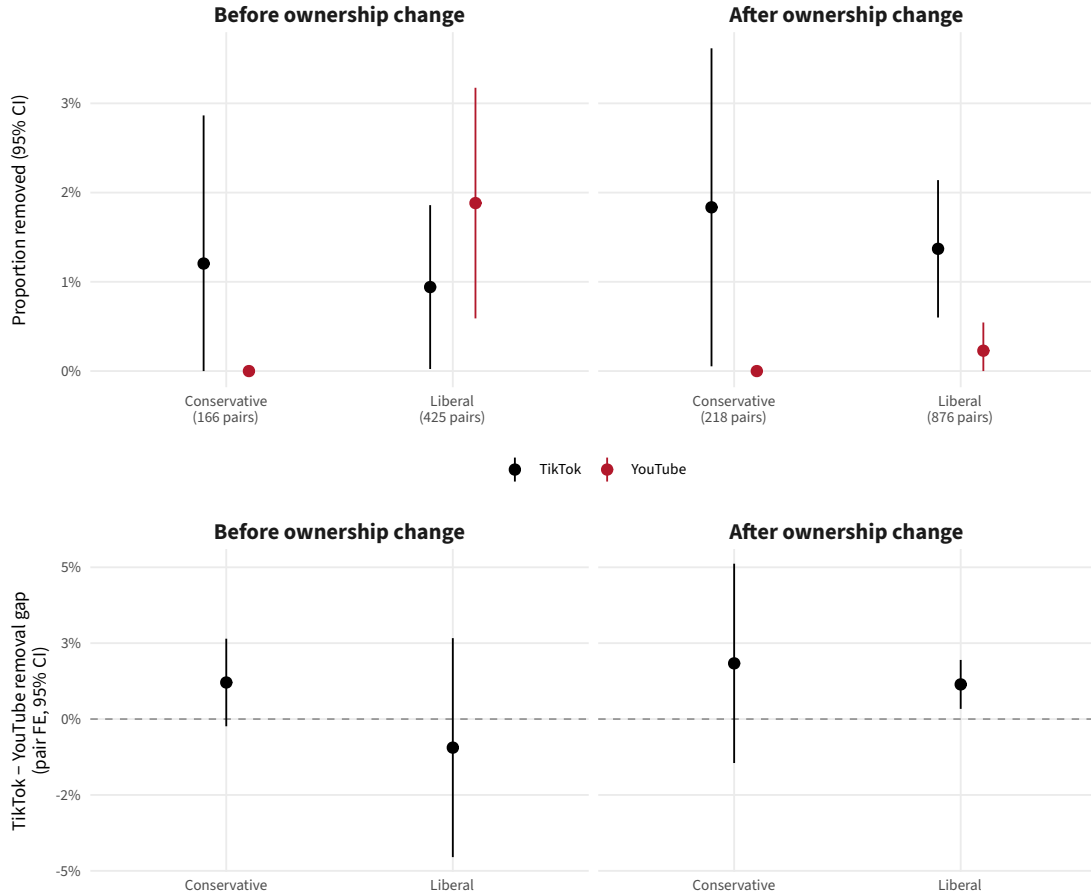


Figure 13: Removal of matched cross-posted videos on TikTok vs. YouTube by creator ideology, for pairs posted in the two months before vs. after the ownership change. Top: proportion removed with 95% confidence intervals. Bottom: within-pair TikTok–YouTube removal gap from a pair fixed-effects regression, with creator-clustered 95% confidence intervals. Although the bottom-panel point estimates switch ordering across periods, the conservative–liberal difference in the gap is not distinguishable from zero in either period. Pairs are restricted to cross-posts within seven days of each other.

Demotion Results: China-related posts

Looking across content categories in Figure 14, we see that demotion is on average less likely for hateful than not hateful videos, especially when measured with the Gompertz model. This may reflect an issue of selection on the dependent variable linked to the removal results presented above. If hateful content is more likely to be removed, the unremoved hateful content may be less likely to be demoted. Put differently, had the removed hateful content not been removed, it would probably have been demoted. Across both models, demotion is significantly associated content categorized as news clips, which could indicate that demotion is used for videos that the algorithm flags as potentially containing intellectual property violations. While China-related videos are slightly more likely to be removed, they are significantly less likely to be demoted. We do not see a lot of association between stance toward China and demotion, only one coefficient (pro-China tech) is significantly less associated with demotion, while overall pro-China stance is somewhat more associated with demotion. These patterns generally hold in a multivariate regression and with author fixed effects (Figure 15).

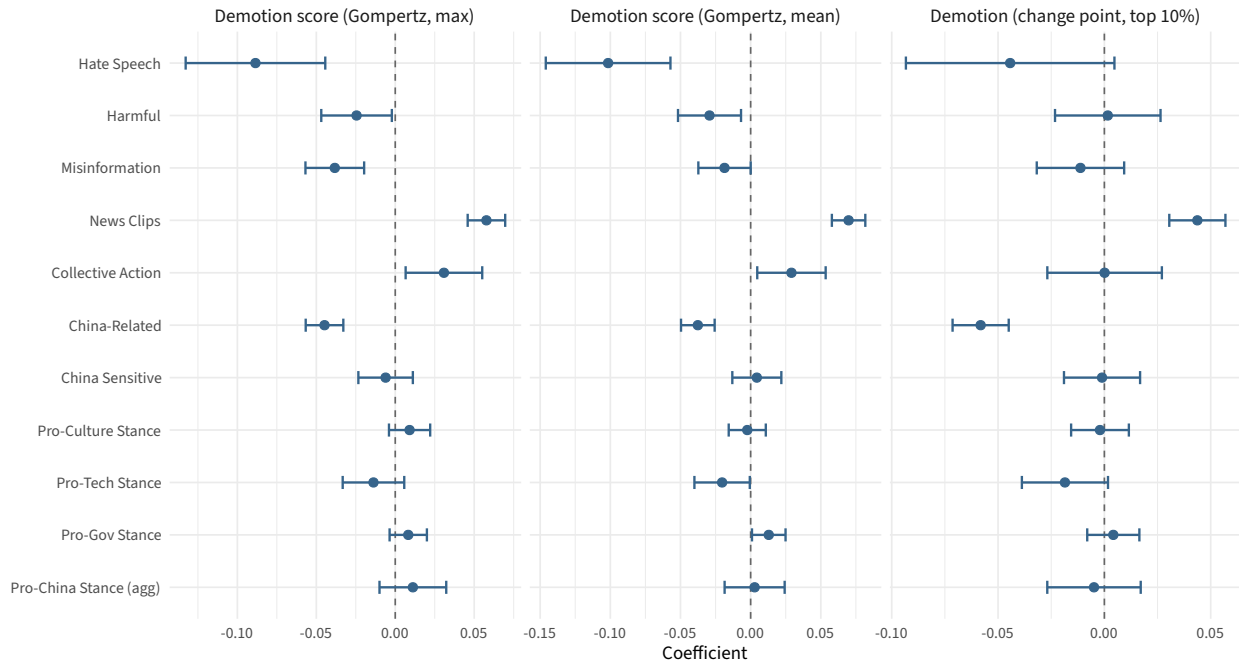


Figure 14: Bivariate associations between Gemini content labels and demotion in the China sample. Each row within a panel is a separate regression of the demotion outcome on a single label; points are OLS coefficients with 95% confidence intervals. Outcomes are the two continuous Gompertz demotion scores (percentile ranks of the maximum and mean observed-minus-predicted gap) and the change point demotion indicator of whether the change point is in the top-10% selectivity cutoff.

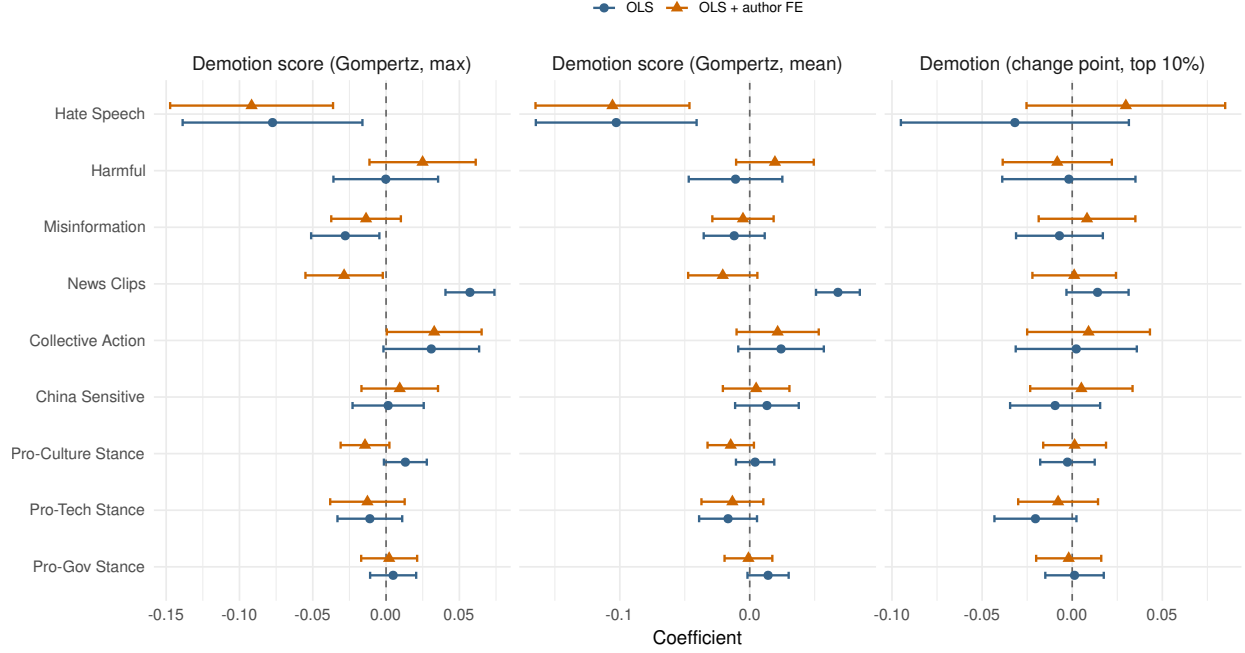


Figure 15: Multivariate associations between Gemini content labels and demotion in the China sample, estimated among China-related videos only. All labels enter jointly, with the three community-guidelines flags (hate speech, harmful, misinformation) entered separately and stance entering as its three dimensions. Blue circles: OLS; orange triangles: OLS with author fixed effects (within-creator comparisons). Bars are 95% confidence intervals.

How does demotion change relative to matched YouTube accounts? We measure YouTube demotion by applying the same two models to YouTube view trajectories, with two adaptations to YouTube’s hourly (rather than 30-minute) snapshot cadence: the survival cutoff for the Gompertz model is set at 71.0 rather than 71.5 hours, and the change point detector anchors at the first observation within three hours of posting (carried backward) rather than requiring a first-hour observation. As on TikTok, each video’s demotion score is converted to a percentile rank within its own platform’s full sample. Because these are within-platform constructs, demotion “levels” are not comparable across platforms by design; only the platform $\times$ group interactions — whether a group sits relatively deeper in TikTok’s demotion distribution than in YouTube’s — are identified.

In Figure 16, pooling pre- and post-ownership periods, we find that YouTube is more likely to demote sensitive China videos than TikTok — a pattern that is consistent and statistically precise across demotion models and cutoffs. The pro-CCP stance interactions point in the direction of TikTok demoting pro-CCP videos relatively *more* than YouTube at most cutoffs, but these stance contrasts are imprecise. Both patterns run contrary to expectations that TikTok would have a more pro-CCP tilt, and could reflect the pressure that TikTok felt to not be pro-CCP.

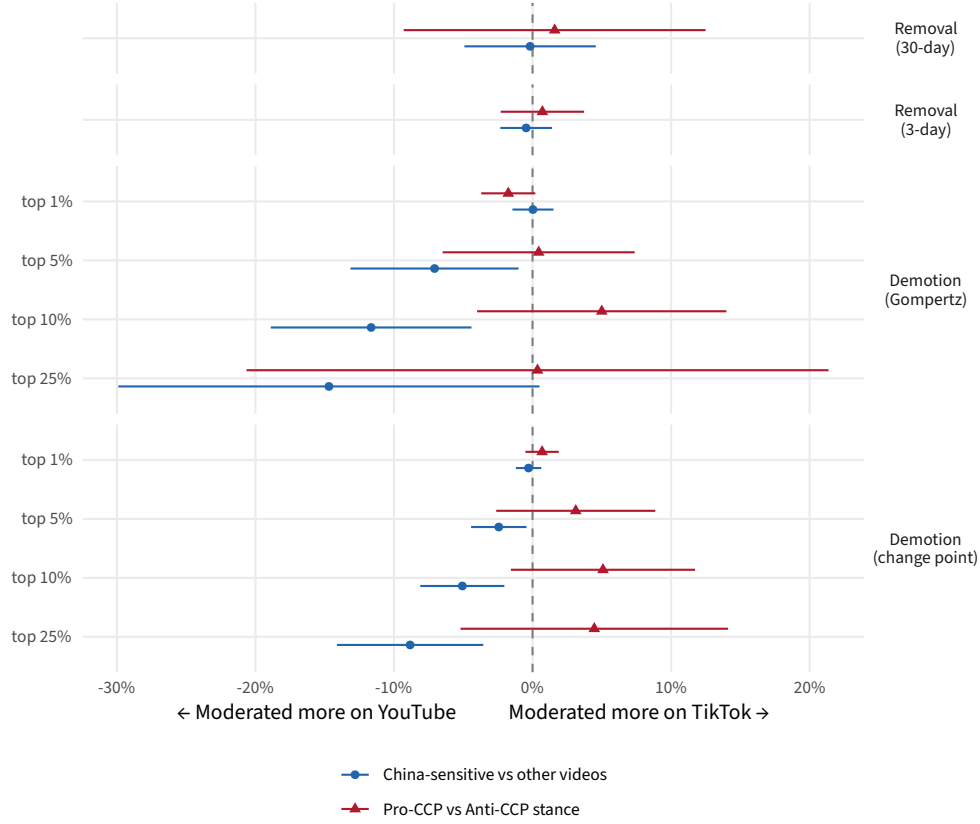


Figure 16: TikTok-YouTube differences in removal and demotion for matched pairs in China sample, pooling the pre- and post-takeover periods. Confidence intervals are clustered by creator. Red triangles: pro-CCP vs. anti-CCP stance (1,212 removal / 1,032 demotion pairs); blue circles: China-sensitive vs. other labeled videos (9,505 / 7,836 pairs). Positive values mean the group is flagged relatively more on TikTok than on YouTube.

Finally, we ask whether the pro-CCP demotion differential changed at the ownership change, estimating the same triple-difference specification used for removal (Equation 7, with the pro-CCP stance indicator in place of the Conservative indicator). Figure 17 reports the platform $\times$ pro-CCP $\times$ post-takeover interaction across both demotion models and all selectivity cutoffs (≈ 970 stance-labeled demotion pairs). Every estimate is statistically indistinguishable from zero, and the signs flip across models and cutoffs: we find no evidence that the ownership change shifted the relative demotion of pro- vs. anti-CCP content on TikTok. The estimates are imprecise, however — with only 237 anti-CCP pairs, the confidence intervals admit large effects in either direction, so this is best read as an absence of evidence rather than a precisely estimated null.

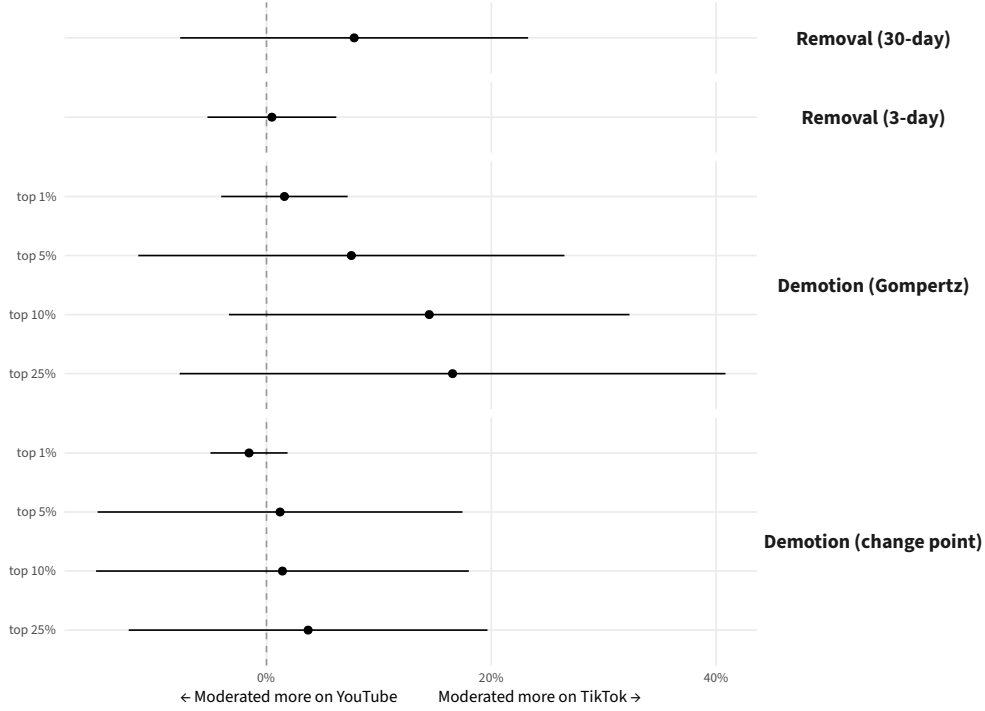


Figure 17: Change in the pro-CCP cross-platform differential after the ownership change (China matched pairs): the platform $\times$ pro-CCP $\times$ post-takeover triple interaction from pair fixed-effects regressions, with creator-clustered 95% confidence intervals. Positive values mean that after January 22, 2026, pro-CCP content is flagged relatively more on TikTok (vs. YouTube) than before, compared with anti-CCP content.

Demotion Results: U.S.-related posts

Similar to the China case, we do not see significant shifts in demotion by stance pre and post the ownership change for the U.S. accounts, regardless of method for estimating demotion. Figure 18 shows demotion by stance over time by estimation method and threshold. Overall, liberal accounts in our sample are slightly more likely to be demoted, but this does not seem to change significantly before versus after the ownership change.

We then examine matched videos in the U.S. sample between YouTube and TikTok. In Figure 19, we show a pooled estimate of relative demotion of conservative content on TikTok relative to YouTube. Overall, we see relatively few effects besides a very small estimate at the 1% threshold that would indicate that TikTok demotes liberal content slightly more relative to YouTube (-1.5 percentage points for conservatives, $p = 0.03$ — threshold-fragile, as the looser cutoffs are all null).

Last, we estimate a triple difference estimator of TikTok by ideology by ownership takeover. Here we see null results except for the Gompertz model at the 25% threshold and the continuous early-peaking score,

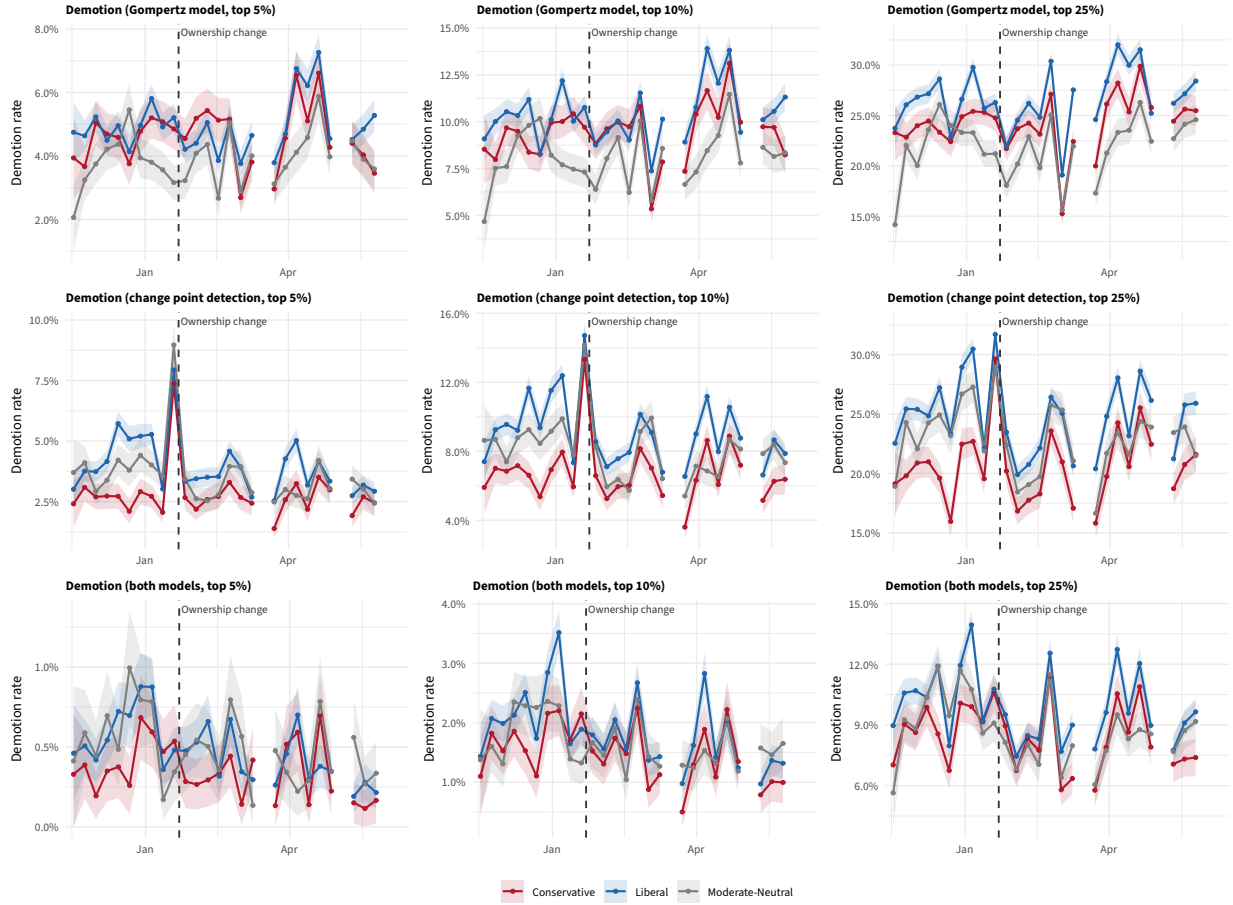


Figure 18: Weekly demotion rates in the U.S. sample by account ideology (liberal, neutral, conservative), around the ownership change (dashed vertical line), across demotion models (rows: Gompertz, change point detection, agreement of both models) and selectivity cutoffs (columns: top 5%, 10%, 25%). Points are weekly rates with 95% Wilson confidence-intervals.

which would indicate that TikTok is more likely to demote Conservative content after the ownership change. However, this result is not robust to threshold change — the policy-relevant strict cutoffs (top 1–10%) are all null — so we do not put much stake into it.

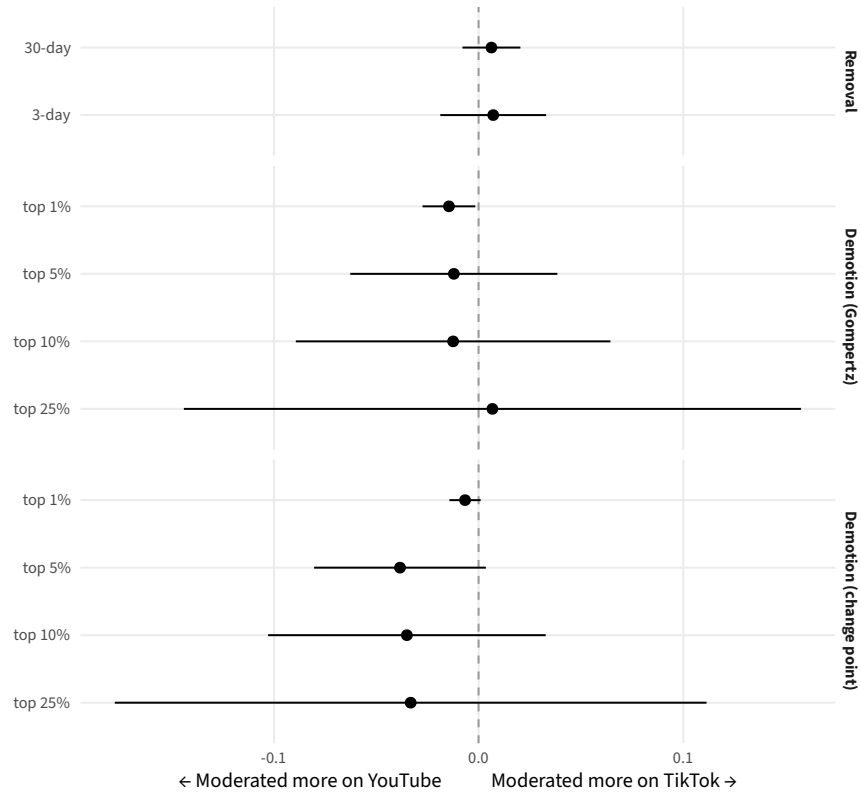


Figure 19: TikTok vs. YouTube differences in removal and demotion for matched U.S. pairs, pooling the pre- and post-takeover periods. Positive values mean conservative creators' videos are flagged relatively more on TikTok than on YouTube, compared with liberal creators.

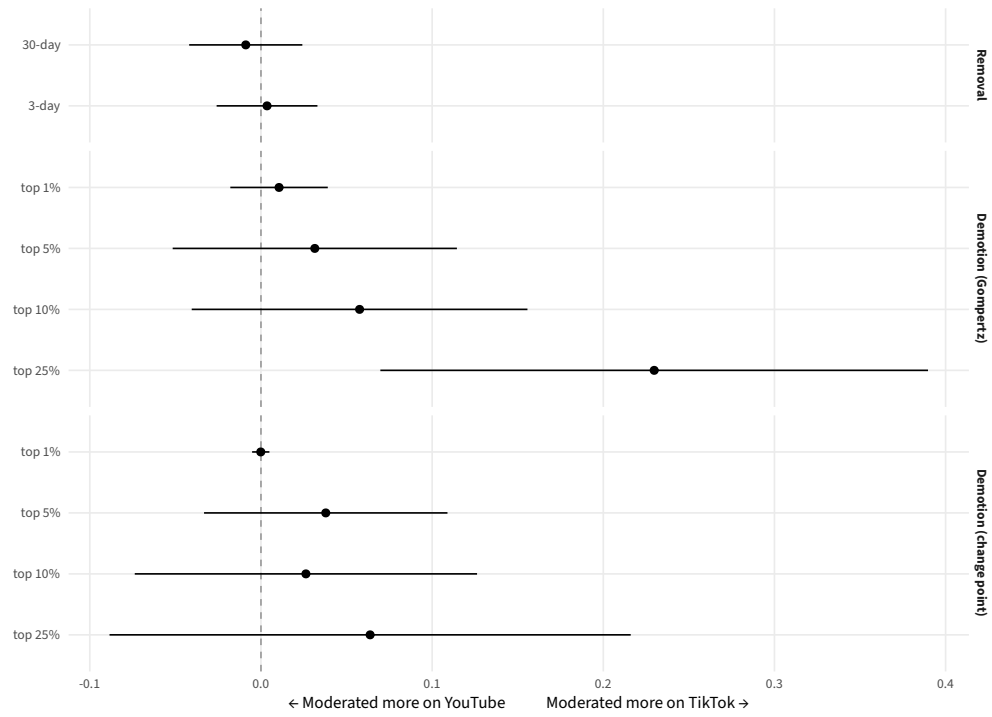


Figure 20: Change in the Conservative cross-platform demotion differential after the ownership change (U.S. matched pairs): the platform $\times$ Conservative $\times$ post-takeover triple interaction from pair fixed-effects regressions with creator-clustered 95% confidence intervals. Positive values mean that after January 22, 2026, conservative creators' content is flagged relatively more on TikTok (vs. YouTube) than before, compared with liberal creators.

Differential Reach at Three Days

Last, we examine two *continuous* reach outcomes: total views at three days, taken from the 30-minute tracking data as the cumulative view count at 72 hours after posting, and views per follower at three days, which divides that count by the creator’s follower count at posting time (using the last observed follower count when the post value is missing or zero). Views per follower is the fairer reach comparison across content types, since total views conflate how widely the algorithm distributes a video with how large the posting account following is. Views are heavily right-skewed, so all inference is on $\log_{10}(1 + x)$. A coefficient β on total views corresponds approximately to a 10^β -fold relative change.

Reach Levels by Content Category

First, we pool both before and after ownership change periods to look at reach levels by content category. Figure 21 predicts reach using OLS regression by content category.

In the China sample, the total-views panel mostly reflects composition: China-related videos and hate-speech videos gather far fewer views, while news clips gather far more. The views-per-follower panel is the more informative one, since it nets out account size: here the three pro-China stance dimensions are each associated with *lower* reach per follower, while China-sensitive content reaches slightly *more* viewers per follower, as does content labeled as misinformation. These signs run against the hypothesis that TikTok’s distribution algorithm favored pro-CCP or suppressed China-sensitive content — consistent with the demotion-flag results in the main text.

In the U.S. sample, hate-speech videos reach fewer viewers, whereas videos labeled harmful-but-not-hate reach more. Conservative political videos gather about 17% fewer total views than liberal ones, but the per-follower difference is null.

Reach Around the Ownership Change

Figures 23–25 show mean reach by group and period for the two months before vs. after the ownership change. Table 4 reports the corresponding difference-in-differences estimates, $\log_{10}(1 + y) \sim \text{Post} \times \text{Group}$ with standard errors clustered by creator.

The ownership change moved none of these reach outcomes differentially: all four $\text{Post} \times \text{Group}$ interactions are statistically indistinguishable from zero.

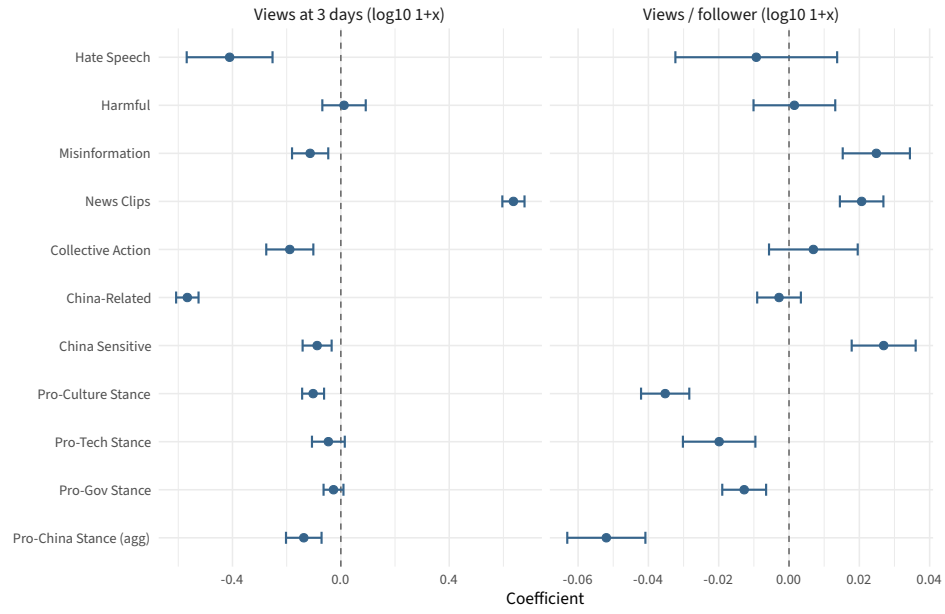


Figure 21: Bivariate associations between Gemini content labels and reach at three days, on $\log_{10}(1+x)$ (left: total views; right: views per follower). Stance variables are coded $-1/0/1$ (anti/neutral/pro).

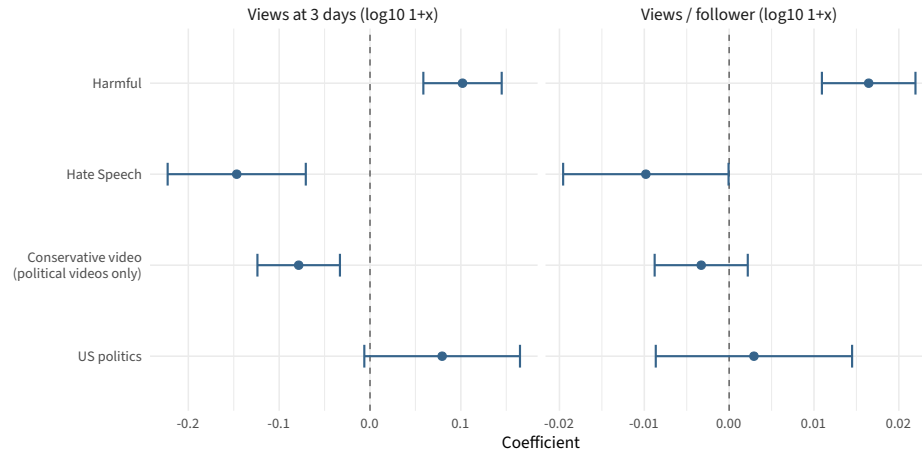


Figure 22: U.S. sample: bivariate associations between the four available Gemini labels and reach at three days, on $\log_{10}(1+x)$. “Conservative video” is defined among U.S.-politics videos only (conservative = 1, liberal = 0). One OLS regression per label; 95% confidence intervals.

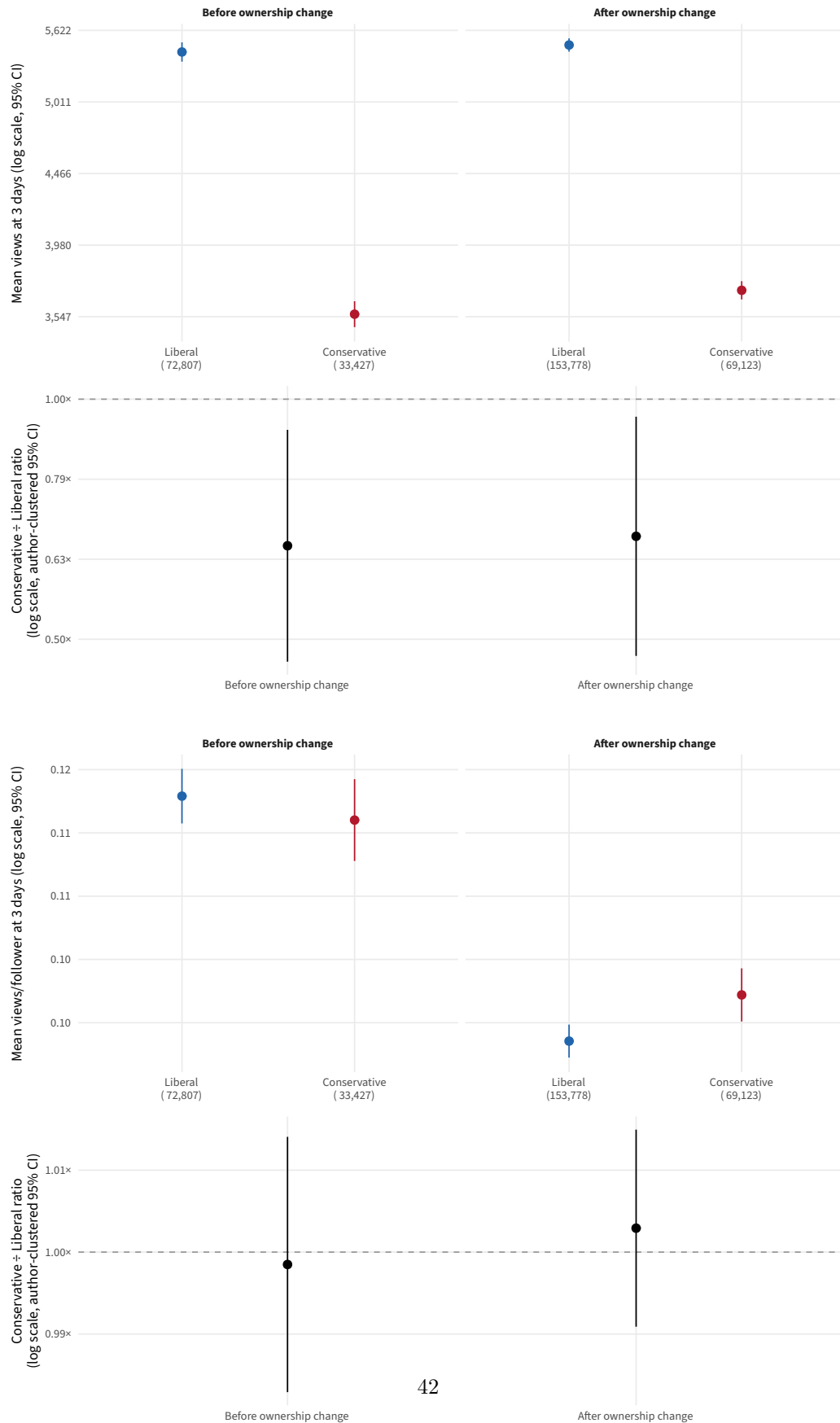


Figure 23: U.S. sample: reach at three days by account ideology, before vs. after the ownership change. Top pair of panels: total views at 3 days; bottom pair: views per follower. Within each: group means of $\log_{10}(1 + x)$ with 95% confidence intervals (upper panel, geometric-mean scale) and the within-period Conservative÷Liberal ratio with creator-clustered 95% confidence intervals (lower panel); the change in that ratio across periods is the difference-in-

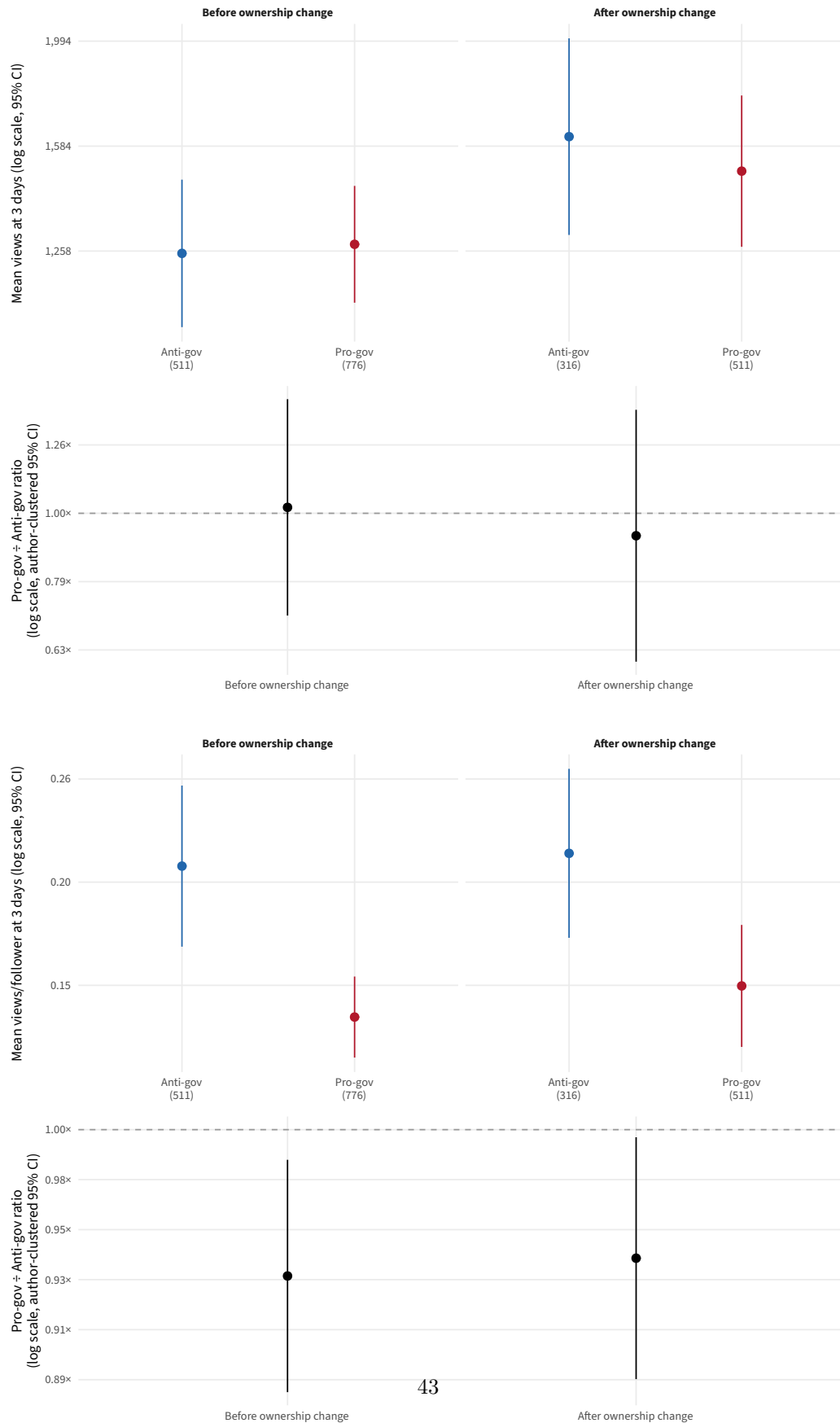


Figure 24: China sample: reach at three days by video stance toward the CCP (Gemini labels), before vs. after the ownership change. Construction as in Figure 23, with Pro-gov vs. Anti-gov stance as the focal contrast.

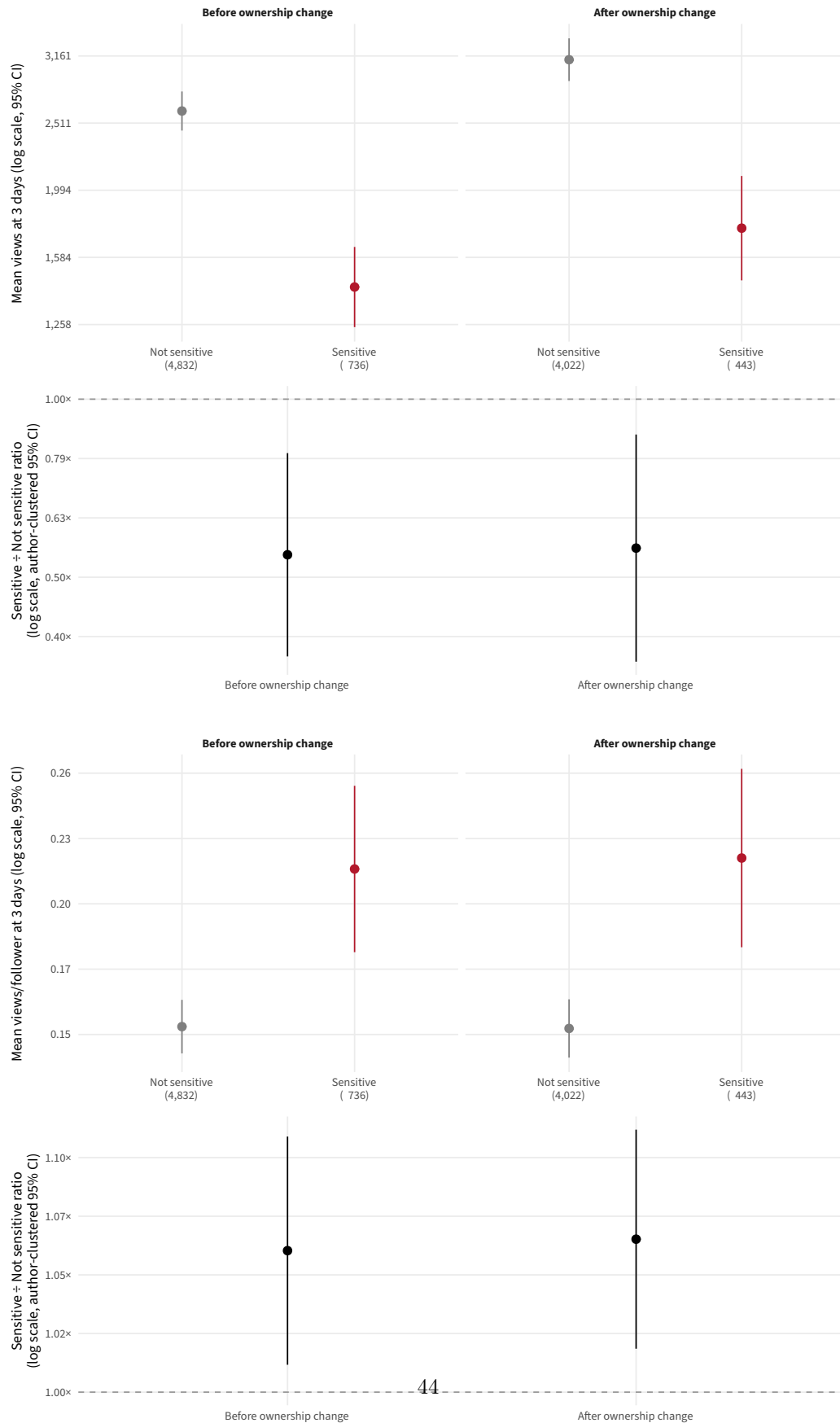


Figure 25: China sample: reach at three days by China-sensitivity (Gemini labels), before vs. after the ownership change. Construction as in Figure 23, with Sensitive vs. Not-sensitive as the focal contrast.

Table 4: Reach around the ownership change: difference-in-differences on $\log_{10}(1+x)$. “Group” is Conservative creators (US, account ideology) and pro-CCP stance videos (China, Gemini labels). DV in columns (1)–(2) is total views at 3 days; columns (3)–(4) views per follower at 3 days. The Post $\times$ Group interaction is the DiD; standard errors clustered by creator in parentheses. A coefficient β means a 10^β -fold relative change. Scrape-gap cohorts dropped.

	Total views (3d)		Views/follower (3d)	
	(1) US	(2) China	(3) US	(4) China
Post (after ownership change)	0.005 (0.017)	0.111 (0.064)	-0.008*** (0.001)	0.002 (0.012)
Group (focal)	-0.183* (0.074)	0.009 (0.081)	-0.001 (0.004)	-0.029* (0.012)
Post $\times$ Group (DiD)	0.012 (0.029)	-0.042 (0.082)	0.002 (0.003)	0.004 (0.015)
Observations	329,135	2,114	329,135	2,114
Creators	1,273	441	1,273	441
Author clustering	Yes	Yes	Yes	Yes

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. All standard errors clustered by creator.

Conclusion

In sum, our paper presents four complementary methodological advances to help audit platform behavior, measuring keyword blocking, content removal, algorithmic demotion, and differential reach. We apply these to one of the most politically salient platform governance events of the past decade: the January 2026 transfer of U.S. TikTok from China-based ByteDance to a U.S.-controlled joint venture. Across all four methods and both content domains, we find no substantial evidence that the ownership change shifted how TikTok moderated China-related or U.S. political content.

The absence of a detectable effect is in and of itself an important finding. Concerns about TikTok’s moderation are in part what motivated years of Congressional action and ultimately legislation. Our results, covering both removal and the more elusive phenomenon of algorithmic demotion, provide one of the first large-scale empirical tests of these concerns and find no systematic evidence that the ownership change has had a measurable effect on moderation in the U.S. or China political domains.

Perhaps the most striking finding of our analysis is that overall, TikTok seems to moderate more pro-China content in the pre-ownership change period, both relative to anti-China content and relative to YouTube. We find that pro-China content was removed at demoted at somewhat higher rates than anti-China content on TikTok throughout our study window, and that YouTube demotes China-sensitive content relatively more than TikTok. This is opposite what the dominant hypothesis would predict. The views-per-follower reach analysis reinforces this pattern: anti-China and China-sensitive content reaches more views per follower, not fewer, than pro-China content on TikTok. One possibility is that TikTok, operating under both Congressional and public scrutiny during our data collection period, adjusted its moderation to be visibly non-preferential toward pro-China content. We emphasize that this finding does not rule out that TikTok was more favorable to pro-China content in earlier time periods before our data collection started.

We note several limitations. Most importantly, our removal measure cannot distinguish platform-initiated takedowns from creator-initiated deletions. Similarly, our two demotion measures – the Gompertz model and the change-point detection algorithm – identify different populations of videos and lack ground truth external validation. We treat them as proxies for demotion rather than confirmed measures of it. The China stance analysis in the YouTube-TikTok matched pairs is additionally limited in statistical power, we only have 237 anti-CCP matched pairs. The null results for that subsample are therefore somewhat uninformative and should not be taken as a precise zero.

Beyond the TikTok application, the methods we develop address a problem in platform accountability:

how to audit moderation behavior when platforms do not share data and when algorithmic demotion is difficult to detect. The cross-platform matched-pairs design, in which identical videos posted by the same creators on both TikTok and YouTube are tracked in parallel, provides a content-controlled comparison. The Gompertz and change-point models are new methods for operationalizing demotion through view trajectories. Both approaches could be replicated on any platform where video metadata can be collected with high frequency.

Platform auditing like what we do here is necessarily ongoing rather than final. Ownership structures, content moderation teams, algorithms, and political environments are constantly evolving. Thus, our snapshot audit of a single event is just that – and cannot capture platform behavior over time. As our next step, we plan to extend our collection window, refine our content labeling approach, and develop further external validation for our demotion estimators. We plan to continue to track the TikTok ownership transition, as there is the possibility that moderation changes are made at a delay as public scrutiny subsides.

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Selection into the Matched Sample

Because only a subset of each account’s videos are cross-posted and successfully matched across platforms, we check whether the probability that a TikTok video is matched to a YouTube counterpart varies with the grouping variables used in the cross-platform analysis. For each sample we take every TikTok video from an account that has a matched YouTube channel and regress an indicator for being matched on the posting account’s group (linear probability model, standard errors clustered by account). In the U.S. sample match propensity is flat across the ideology spectrum (Figure 26). In the China sample the raw match rate rises with pro-CCP stance (pro-CCP 10.5%, neutral 5.8%, anti-CCP 1.6%), though with account-clustered standard errors the group differences are imprecise (Figure 27).

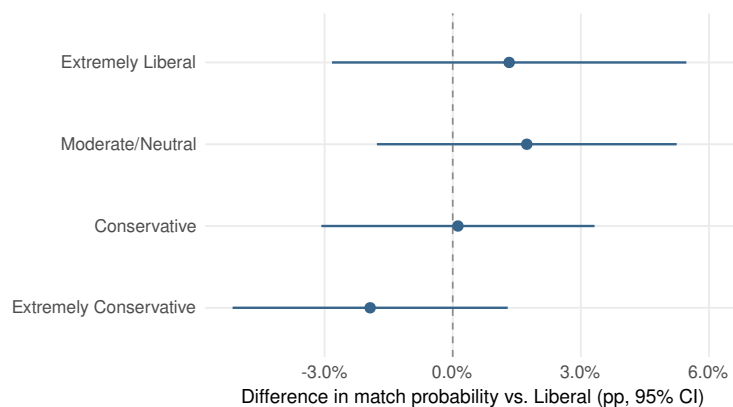


Figure 26: U.S. sample: difference in the probability that a TikTok video is matched to a YouTube counterpart, by the posting account’s ideology (relative to Liberal). Linear probability model over all TikTok videos from accounts with a matched YouTube channel; points are percentage-point differences with account-clustered 95% confidence intervals.

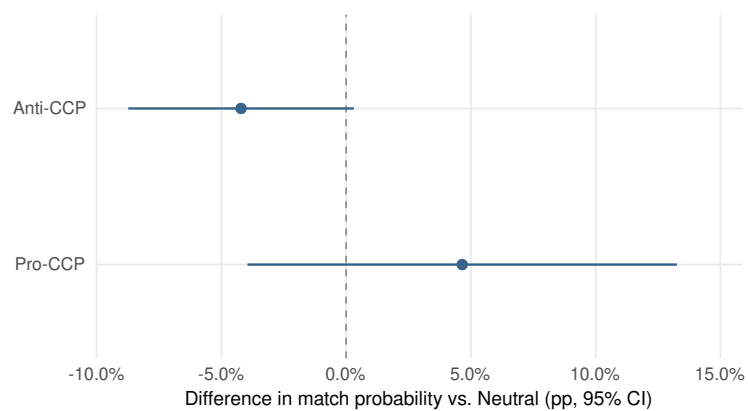


Figure 27: China sample: difference in the probability that a TikTok video is matched to a YouTube counterpart, by the posting account’s stance toward the CCP (account-level stance, relative to Neutral). Linear probability model over all TikTok videos from accounts with a matched YouTube channel; points are percentage-point differences with account-clustered 95% confidence intervals.

Removal Analysis Robustness Check and Event Studies

Robustness: Removal at the 3-Day Horizon

Our main cross-platform removal measure reads YouTube removal over a 30-day window (a video- or whole-channel suspension within 30 days of the YouTube post), horizon-matched to TikTok's 30-day status re-check. As a shorter-horizon check we re-measure both platforms at 3 days: TikTok's gap-robust 3-day measure against a YouTube suspension within 3 days of the post. Matched-sample YouTube suspensions are rare at this short horizon (no matched U.S. video, and only a handful of China videos, were suspended within 3 days), so the 3-day gap is driven largely by TikTok removals; the stance, sensitivity, and ideology contrasts remain null.

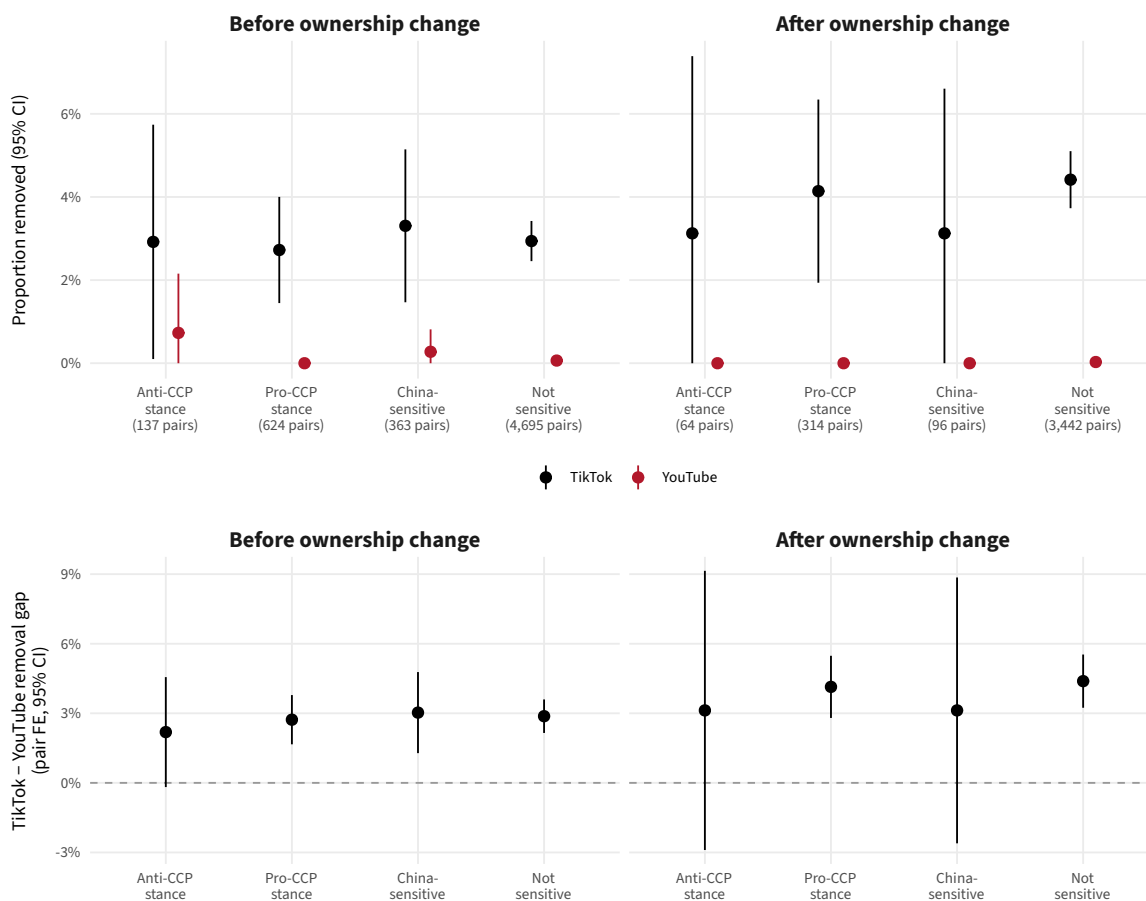


Figure 28: Robustness: removal of matched cross-posted China-sample videos at the 3-day horizon (TikTok's gap-robust 3-day measure vs. YouTube suspension within 3 days of posting), the shorter-horizon companion to Figure 10 (30-day). The stance and sensitivity contrasts remain null.

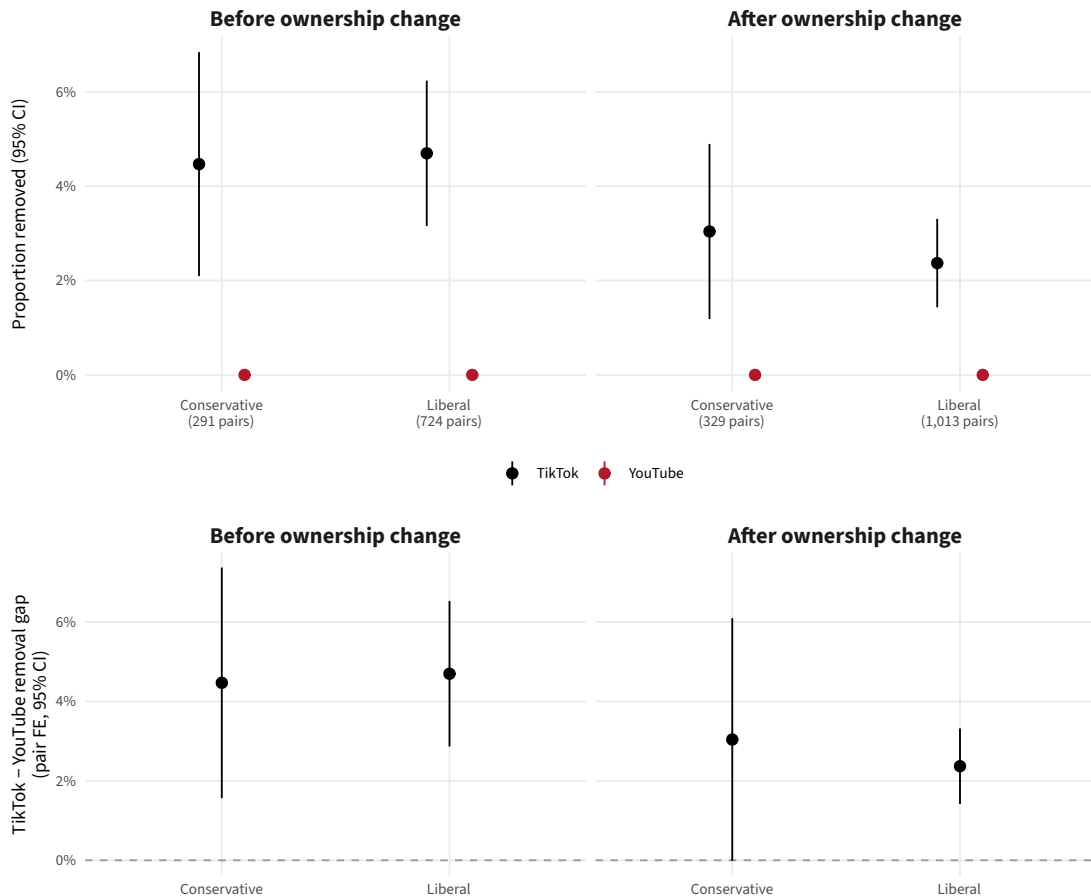


Figure 29: Robustness: removal of matched cross-posted U.S. videos at the 3-day horizon (TikTok’s gap-robust 3-day measure vs. YouTube suspension within 3 days of posting), the shorter-horizon companion to Figure 13 (30-day). The overall TikTok–YouTube gap is larger at 3 days — YouTube suspensions are rare that quickly, so it mostly reflects TikTok removals — and the conservative differential is null at both horizons.

Event Studies and Parallel Trends

The difference-in-differences designs above assume that, absent the ownership change, the within-pair TikTok–YouTube gap — and, for the triple difference, its group differential — would have evolved in parallel. We probe this directly with event studies: binning matched pairs by their TikTok posting cohort (biweekly; monthly for the thin China stance subset), we estimate the cross-platform gap in each cohort relative to the bin just before January 22, 2026, replacing the single $TT \times Post$ term in Equation 7 with a full set of cohort $\times$ platform interactions (pair fixed effects and creator-clustered standard errors as before). A second specification adds cohort $\times$ platform $\times$ group interactions, whose pre-takeover coefficients test the parallel-trends assumption of the *triple-difference* estimand directly. For each series we report a joint Wald

Table 5: Matched YouTube–TikTok removal (3-day proxy, robustness): within-pair platform differences by group. Companion to the 30-day status-check table (Table 3). “Group” is the focal creator/content group per sample: Conservative creators (US, account ideology) and pro-CCP stance videos (China, Gemini labels on all matched videos). Columns (1)/(3): pair fixed-effects difference-in-differences; columns (2)/(4): triple difference adding the post-takeover interaction. All columns restrict to pairs cross-posted within seven days; TikTok scrape-gap cohorts dropped throughout; standard errors clustered by creator in parentheses. TikTok removal is the gap-robust 3-day measure; YouTube removal is point-in-time unavailability (late-March 2026 pull for the US sample, ~June 2026 for China), so post-takeover cohorts — especially in the US — have had mechanically less YouTube exposure (a horizon mismatch the 30-day table avoids). The large China Post terms in column (4) are driven by a single creator whose YouTube channel became unavailable after the takeover; the creator-clustered standard errors reflect this.

	US (Group = Conservative)		China (Group = Pro-CCP)	
	(1) DiD	(2) Triple diff.	(3) DiD	(4) Triple diff.
TikTok	0.030*** (0.004)	0.040*** (0.008)	0.025 (0.014)	0.022 (0.012)
TikTok × Group	0.007 (0.013)	0.003 (0.017)	0.007 (0.015)	0.005 (0.013)
TikTok × Post		-0.014 (0.007)		0.009 (0.029)
TikTok × Group × Post		0.004 (0.015)		0.005 (0.029)
Observations	8,980	8,980	2,278	2,278
Matched pairs	4,490	4,490	1,139	1,139
Pair fixed effects	Yes	Yes	Yes	Yes

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

test that all pre-takeover coefficients equal zero.

Figures 30 and 31 show the results. The identifying variation for our group comparisons is clean: in the U.S. sample, both the pooled gap and the Conservative differential are flat before the takeover for removal and Gompertz demotion (joint $p = 0.43$ – 0.94), and no post-takeover cohort departs from the pre-period. In the China sample the *group differentials* — the estimand of every stance and sensitivity comparison in this paper — likewise show no pre-trends for the headline measures (China-sensitive: joint $p \geq 0.52$ on all four outcomes; pro-CCP: $p = 0.73$ for 30-day removal and $p = 0.32$ for Gompertz demotion). Two caveats emerge. First, the China removal event study runs on the sparse 3-day horizon (YouTube suspensions within 3 days of posting are rare), where both the pooled gap and the stance and sensitivity differentials fail the pre-trend test ($p \leq 0.02$); we therefore read the China removal evidence from the pooled 30-day estimates (Table 3) rather than the temporal trend. Second, the change-point demotion outcome shows pre-trend violations in several series (U.S. pooled gap $p = 0.001$; China pro-CCP differential $p < 0.001$), suggesting cohort-varying behavior of that detector; we accordingly place more weight on the Gompertz demotion and the 30-day removal results, and continue to treat the China stance contrasts as exploratory.

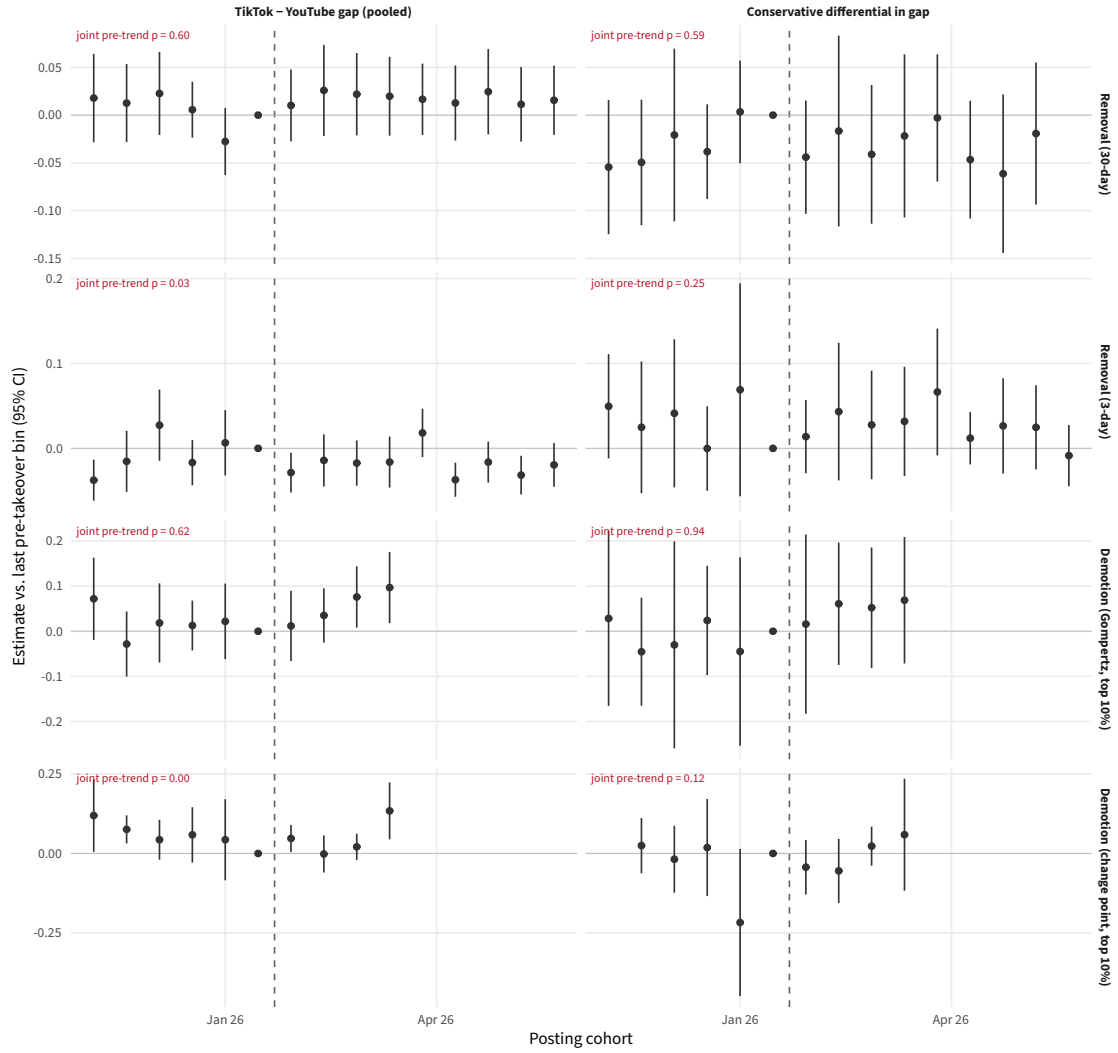


Figure 30: Event-study estimates for the U.S. matched sample: the within-pair TikTok–YouTube gap (left column) and the Conservative–Liberal differential in that gap (right column) by biweekly posting cohort, relative to the last pre-takeover bin (dashed line; pair fixed effects, creator-clustered 95% confidence intervals; pairs cross-posted within seven days). Red annotations report joint Wald tests that all pre-takeover coefficients are zero. TikTok removal is the 30-day status check (top row) and the gap-robust 3-day measure (second row); YouTube removal is the horizon-matched suspension measure (within-30-day and within-3-day of posting). Demotion outcomes are fixed-72-hour percentile ranks and are cohort-clean. Bins with too few treated observations are hidden.

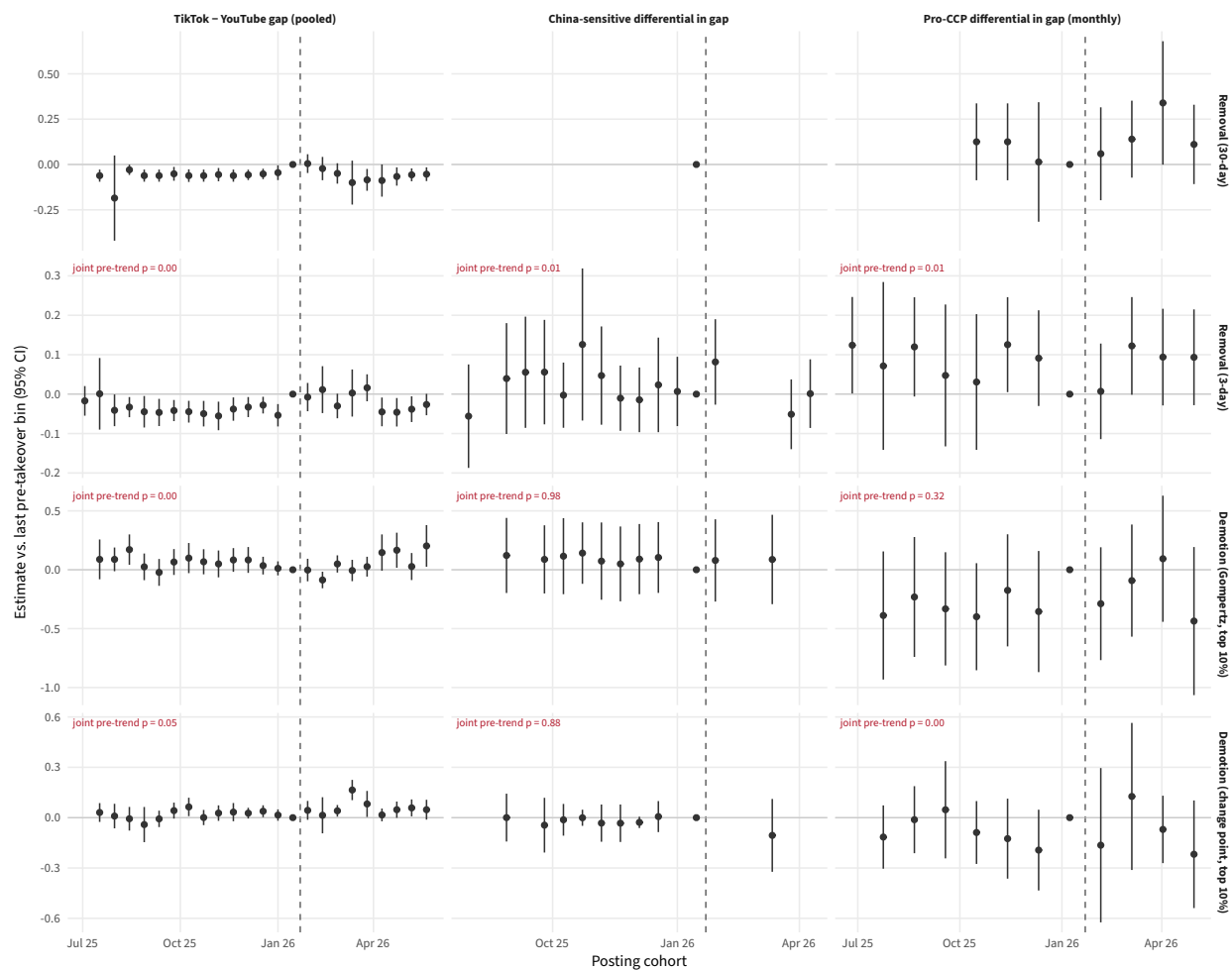


Figure 31: Event-study estimates for the China matched sample: the pooled within-pair TikTok–YouTube gap (left), the China-sensitive differential (middle, biweekly), and the pro-CCP differential (right, monthly given sparse stance cells) by posting cohort, relative to the last pre-takeover bin. Construction as in Figure 30. Joint pre-trend tests are computed over all estimable pre-takeover bins; displayed points are restricted to adequately filled bins.